# Analysis note: charged-particle multi-particle cumulants in ALEPH hadronic $e^+e^-$ collisions at the $Z$ pole

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**Abstract**

This note documents a first charged-particle multi-particle cumulant analysis of archived ALEPH hadronic $e^+e^-$ collisions at $\sqrt{s} \simeq 91.2$ GeV. The analysis follows the public Electron-Positron Alliance note style: the main body defines the observable, data setup, event and track selections, cumulant calculation, statistical uncertainty treatment, and the current preliminary results. The reported observables are $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity, evaluated with respect to both the laboratory beam axis and the event thrust axis. The cumulant implementation uses exact ordered, distinct-particle correlations through eighth order and propagates statistical uncertainties with delete-one-chunk jackknife resampling over sixteen event chunks. The present result is a detector-level charged-particle study; detector-response, acceptance, and model-dependent systematic uncertainties are not yet included. A 1994 reconstructed-MC baseline is included for detector-level comparisons of the beam-axis and thrust-axis trends, including rapidity-gap and two-subevent cumulant constructions.

**Keywords:** ALEPH, archived $e^+e^-$ data, charged-particle cumulants, flow harmonics, Q-cumulants

## Contents

# 1 Introduction

Long-range azimuthal correlations and multi-particle cumulants are central tools for separating collective and non-collective correlation structures in high-energy collisions. In hadronic and nuclear collisions, the hierarchy of $v_n\{2\}$, $v_n\{4\}$, $v_n\{6\}$, and $v_n\{8\}$ provides a compact way to test whether the measured azimuthal anisotropy is dominated by a common event-level geometry or by few-particle correlations. The same techniques can be applied to archived $e^+e^-$ data, where the absence of beam remnants and a well-defined color-singlet initial state provide a clean reference for jet fragmentation, parton-shower, and hadronization effects.

The starting point for this analysis is the ALEPH StudyMult charged-particle workflow and the previous ALEPH two-particle-correlation analysis of archived LEP-I data [1]. The broader Electron-Positron Alliance two-particle-correlation program also includes the LEP-II analysis note and the published long-range near-side correlation result, which used the same beam-axis and thrust-axis correlation framework and reported Fourier coefficients from two-particle correlations [2, 3]. The present note extends that infrastructure from two-particle Fourier coefficients to exact multi-particle cumulants through eight particles. The implementation follows the Q-cumulant formalism of Refs. [4, 5] and the cumulant combinations used in small-system flow measurements, including the CMS high-multiplicity pp reference analysis [6].

This document intentionally follows the compact public-note format used in recent Electron-Positron Alliance analysis notes, including the public LEP-II two-particle-correlation note [2]. The main body contains the observable definition, active data setup, analysis method, validation, and preliminary results, while detailed corrections and systematic variations are identified as future work rather than folded into the present detector-level result.

# 2 Observable Definition

For each selected event, charged particles are represented by their azimuthal angles $\phi_i$ in a chosen coordinate system. The present study uses two coordinate definitions:

- beam axis: $\phi_i$ is the laboratory azimuth around the ALEPH beam axis;
- thrust axis: the event is rotated into a frame defined by the event thrust axis, and $\phi_i$ is the azimuth around that axis.

The harmonic used in this note is $n = 2$. For an event with multiplicity $M$, the ordered distinct-particle $2k$-particle correlation is

$$\langle 2k\rangle_n = \frac{1}{M(M-1)\cdots(M-2k+1)} \sum_{\substack{i_1,\ldots,i_{2k}\\ \text{all distinct}}} \exp\left[in\left(\phi_{i_1} + \cdots + \phi_{i_k} - \phi_{i_{k+1}} - \cdots - \phi_{i_{2k}}\right)\right]. \quad (1)$$

The event-ensemble averages of these quantities are accumulated with their natural ordered-tuple weights. The cumulants are then

$$c_2\{2\} = \langle 2\rangle\,, \quad (2)$$
$$c_2\{4\} = \langle 4\rangle - 2\langle 2\rangle^2\,, \quad (3)$$
$$c_2\{6\} = \langle 6\rangle - 9\langle 4\rangle\langle 2\rangle + 12\langle 2\rangle^3\,, \quad (4)$$
$$c_2\{8\} = \langle 8\rangle - 16\langle 6\rangle\langle 2\rangle - 18\langle 4\rangle^2 + 144\langle 4\rangle\langle 2\rangle^2 - 144\langle 2\rangle^4\,. \quad (5)$$

The corresponding flow-like coefficients are defined as

$$v_2\{2\} = \sqrt{c_2\{2\}}, \quad (6)$$
$$v_2\{4\} = \left[-c_2\{4\}\right]^{1/4}, \quad (7)$$
$$v_2\{6\} = \left[c_2\{6\}/4\right]^{1/6}, \quad (8)$$
$$v_2\{8\} = \left[-c_2\{8\}/33\right]^{1/8}. \quad (9)$$

If the required cumulant sign is not physical for a given bin, the corresponding $v_2\{m\}$ point is suppressed in the plotted result and is stored as zero in the exported CSV table.

# 3 Data Setup and Selections

The baseline LEP1 figures in this note use the merged LEP1 1992–1995 reconstructed data sample

`/mnt/data3/data3/yjlee/StudyMultSamples/ALEPH/LEP1/LEP1Merged_20200611.root`.

This file contains 3,050,610 entries in the tree `t`; all entries pass the current charged-particle selection and multiplicity-bin requirement. The input format is the StudyMult fixed-array layout with `year`, `RunNo`, `EventNo`, `nParticle`, `px`, `py`, `pz`, and `pwflag` branches. The year composition is 551,474 events from 1992, 538,601 from 1993, 1,365,440 from 1994, and 595,095 from 1995. A converted 1991 ROOT sample was not found in the available merged samples, so this production uses the available merged LEP1 statistics from 1992–1995 rather than the complete 1991–1995 data set.

Charged particles are selected using the StudyMult charged-particle convention

$$\texttt{pwflag} \in \{0, 1, 2\}, \tag{10}$$

with $p_{\mathrm{T}} > 0.4$ GeV for both the beam-axis and thrust-axis cumulants. The event thrust axis is computed from reconstructed particle momenta, following the local ALEPH StudyMult convention used by the prototype analyses. The charged-particle multiplicity bins used in this note are

$$[0, 10), [10, 15), [15, 20), [20, 25), [25, 30), [30, 35), [35, 40), [40, \infty).$$

The corresponding event counts are shown in Table 1. All beam-axis and thrust-axis cumulant figures are binned on this laboratory selected charged-particle multiplicity; the thrust-selected multiplicity is retained as a diagnostic but is not used as the horizontal-axis binning in the present production.

Table 1: Selected event counts in the multiplicity bins used for the current detector-level cumulant analysis.

| Multiplicity bin | Events |
|---|---|
| $[0, 10)$ | 241919 |
| $[10, 15)$ | 1046419 |
| $[15, 20)$ | 1132225 |
| $[20, 25)$ | 500481 |
| $[25, 30)$ | 113682 |
| $[30, 35)$ | 14738 |
| $[35, 40)$ | 1095 |
| $[40, \infty)$ | 51 |

# 4 Cumulant Implementation

The analysis code computes the ordered correlations in Eq. (1) using an exact elementary-symmetric polynomial recursion over the event-level phase factors $\exp(in\phi_i)$. This avoids self-correlations by construction and uses the same ordered-particle denominator as the Q-cumulant definition. The recursion is applied for 2, 4, 6, and 8 particles in each multiplicity bin and for both axis definitions.

The implementation was checked in two ways. First, the code contains a `--SelfTest 1` mode that compares the production correlator against brute-force ordered distinct-particle sums for low-multiplicity toy events. Second, the cumulant combinations and $v_2\{m\}$ normalizations were compared against the published Q-cumulant formulae in Refs. [4, 5, 6]. The signs and normalization factors in the cumulant and $v_2\{m\}$ definitions above match those references.

## 5 Statistical Uncertainties

The production job was split into sixteen contiguous event chunks. Each chunk stores independent numerator and denominator sums for every correlator and multiplicity bin. The summary step first merges all chunks to obtain the central value and then recomputes the full nonlinear cumulant chain after removing one chunk at a time. The quoted statistical uncertainty is the delete-one-chunk jackknife estimate

$$\sigma_{\mathrm{JK}} = \sqrt{\frac{N_{\mathrm{chunk}} - 1}{N_{\mathrm{chunk}}} \sum_{j=1}^{N_{\mathrm{chunk}}} \left(x_{(j)} - \bar{x}\right)^2}, \tag{11}$$

where $x_{(j)}$ is the result with chunk $j$ removed. This propagates the nonlinear transformations from raw correlations to cumulants and then to $v_2\{m\}$. If any leave-one sample has the wrong cumulant sign for a real-valued $v_2\{m\}$, the corresponding $v_2\{m\}$ point is suppressed rather than assigned a conditional error from only the sign-valid samples. The procedure assumes the chunks are statistically independent; if the input file ordering tracks detector running conditions, the jackknife may also contain a small run-block component.

## 6 Preliminary Results

Figures 1 and 2 show the current detector-level $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ results versus charged-particle multiplicity. In both coordinate systems the two-particle value is larger than the higher-order values, as expected when few-particle and jet-like correlations contribute. The higher-order beam-axis points remain numerically stable through most bins, while some high-order thrust-axis and high-multiplicity points become sign-unstable or carry large jackknife uncertainties because the event counts and ordered eight-particle statistics are limited in those bins.

Table 2: Numerical beam-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as "–" are sign-invalid or unavailable in the current sample.

| Multiplicity | $v_2\{2\}$ | $v_2\{4\}$ | $v_2\{6\}$ | $v_2\{8\}$ |
|---|---|---|---|---|
| $[0, 10)$ | $0.8121 \pm 0.0004$ | $0.8078 \pm 0.0004$ | $0.8075 \pm 0.0004$ | $0.8075 \pm 0.0004$ |
| $[10, 15)$ | $0.7834 \pm 0.0002$ | $0.7720 \pm 0.0002$ | $0.7711 \pm 0.0002$ | $0.7709 \pm 0.0002$ |
| $[15, 20)$ | $0.7396 \pm 0.0002$ | $0.7183 \pm 0.0002$ | $0.7162 \pm 0.0003$ | $0.7159 \pm 0.0003$ |
| $[20, 25)$ | $0.6808 \pm 0.0002$ | $0.6449 \pm 0.0003$ | $0.6411 \pm 0.0003$ | $0.6404 \pm 0.0003$ |
| $[25, 30)$ | $0.6181 \pm 0.0005$ | $0.5662 \pm 0.0006$ | $0.5602 \pm 0.0007$ | $0.5591 \pm 0.0007$ |
| $[30, 35)$ | $0.5620 \pm 0.0016$ | $0.4952 \pm 0.0022$ | $0.4869 \pm 0.0023$ | $0.4852 \pm 0.0023$ |
| $[35, 40)$ | $0.5124 \pm 0.0077$ | $0.4384 \pm 0.0102$ | $0.4262 \pm 0.0110$ | $0.4233 \pm 0.0113$ |
| $[40, \infty)$ | $0.5174 \pm 0.0255$ | $0.4491 \pm 0.0426$ | $0.4345 \pm 0.0518$ | $0.4302 \pm 0.0570$ |

Table 3: Numerical thrust-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as "–" are sign-invalid or unavailable in the current sample.

| Multiplicity | $v_2\{2\}$ | $v_2\{4\}$ | $v_2\{6\}$ | $v_2\{8\}$ |
|---|---|---|---|---|
| $[0, 10)$ | $0.3367 \pm 0.0016$ | – | – | $0.3510 \pm 0.0074$ |
| $[10, 15)$ | $0.3647 \pm 0.0003$ | – | – | – |
| $[15, 20)$ | $0.3983 \pm 0.0004$ | – | – | – |
| $[20, 25)$ | $0.4173 \pm 0.0003$ | $0.2846 \pm 0.0015$ | – | – |
| $[25, 30)$ | $0.4201 \pm 0.0006$ | $0.3165 \pm 0.0015$ | $0.2819 \pm 0.0037$ | $0.2458 \pm 0.0119$ |
| $[30, 35)$ | $0.4107 \pm 0.0023$ | $0.3168 \pm 0.0037$ | $0.3028 \pm 0.0034$ | $0.2991 \pm 0.0035$ |
| $[35, 40)$ | $0.3990 \pm 0.0080$ | $0.3036 \pm 0.0190$ | $0.2873 \pm 0.0243$ | $0.2847 \pm 0.0258$ |
| $[40, \infty)$ | $0.4242 \pm 0.0275$ | $0.3231 \pm 0.0634$ | – | – |

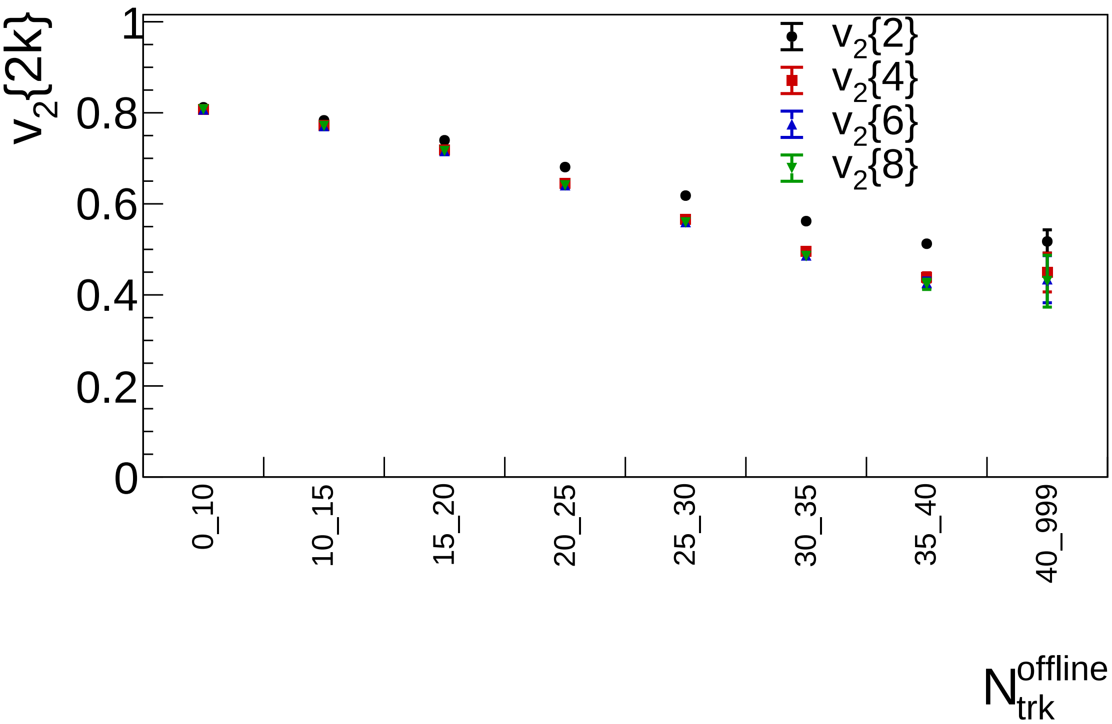



Figure 1: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using the laboratory beam-axis azimuth. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks. Missing high-order points indicate cumulant signs that do not give a real-valued $v_2\{m\}$ in the current sample.

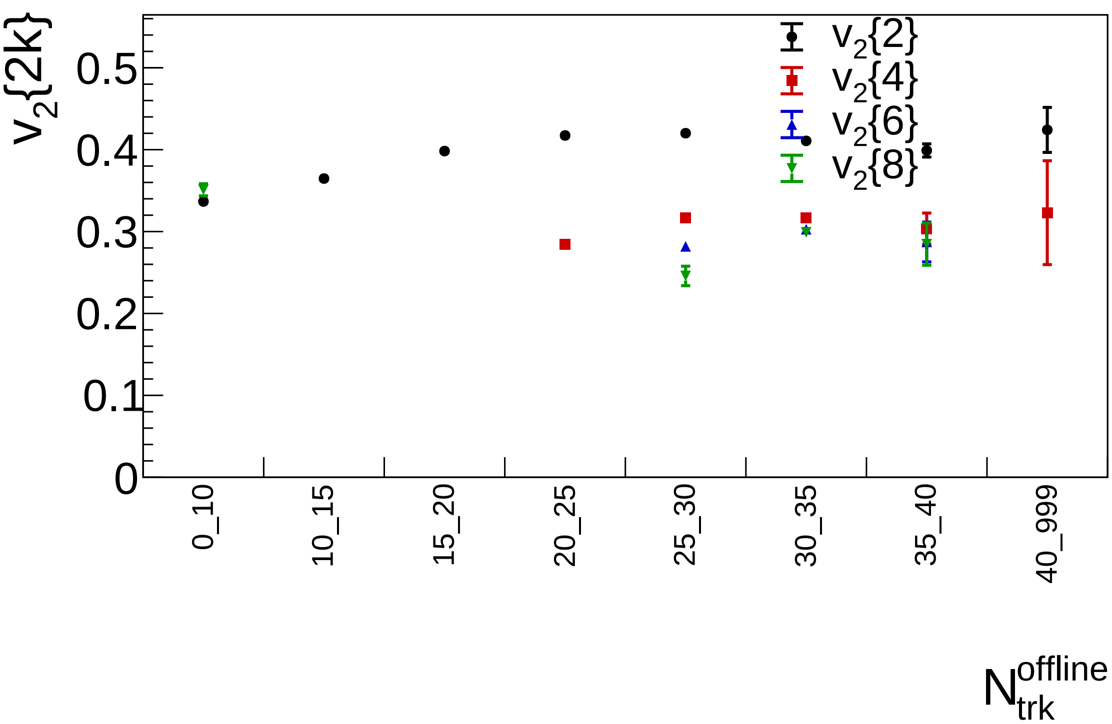



Figure 2: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using azimuth around the event thrust axis. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks.

### 6.1 LEP2 merged-sample data cross-check

The same nominal detector-level cumulant workflow was also run on the requested LEP2 merged sample,

`/mnt/data3/data3/yjlee/StudyMultSamples/ALEPH/LEP2/20180713/LEP2_merged.root`.

The input tree `t` contains 409,270 entries, with year counts 1,983 (1995), 30,914 (1996), 61,076 (1997), 89,685 (1998), 111,493 (1999), and 114,119 (2000). The file-level `isMC` branch is false for all entries in this merged file, so it is treated here as the requested LEP2 data sample. No additional center-of-mass-energy selection is applied in this first pass; the `Energy` branch spans approximately 91–209 GeV. The charged-particle selection, multiplicity bins, beam-axis and thrust-axis definitions, and sixteen-chunk jackknife procedure match the LEP1 nominal production.

Figures 3 and 4 show the LEP2 merged-sample result. The lower LEP2 statistics relative to the LEP1 merged sample lead to visibly larger jackknife uncertainties, especially in the highest multiplicity bins and for higher-order cumulants. The result is kept as a detector-level cross-check until the LEP2 sample composition and energy-window choices are finalized.

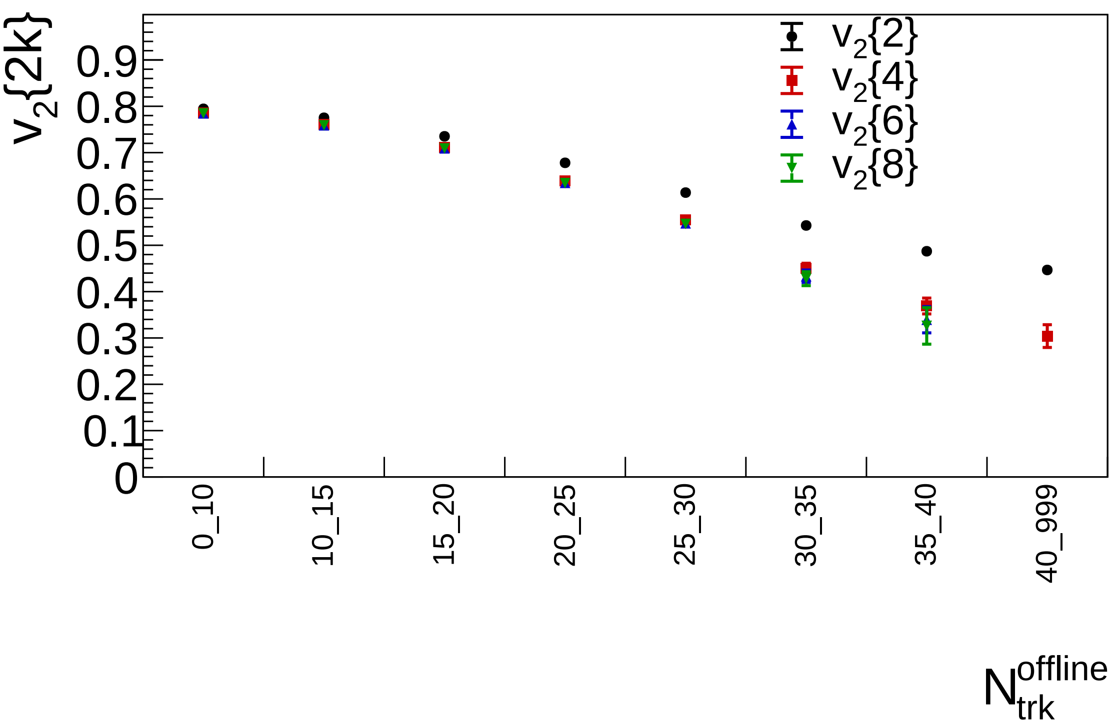



Figure 3: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ in the requested LEP2 merged sample using the laboratory beam-axis azimuth.

For the direct LEP1–LEP2 data comparison, Figures 5 and 6 overlay the merged LEP1 1992–1995 baseline and the requested LEP2 merged sample with the same detector-level selection, multiplicity bins, axis definitions, and jackknife treatment. LEP1 is shown as open circles, while LEP2 is shown as filled circles.

### 6.2 Matched 1994 reconstructed-MC comparison

A separate MC production was run on the available LEP1 1994 reconstructed-MC sample

`/mnt/data3/data3/yjlee/StudyMultSamples/LEP1MCMerged.root`.

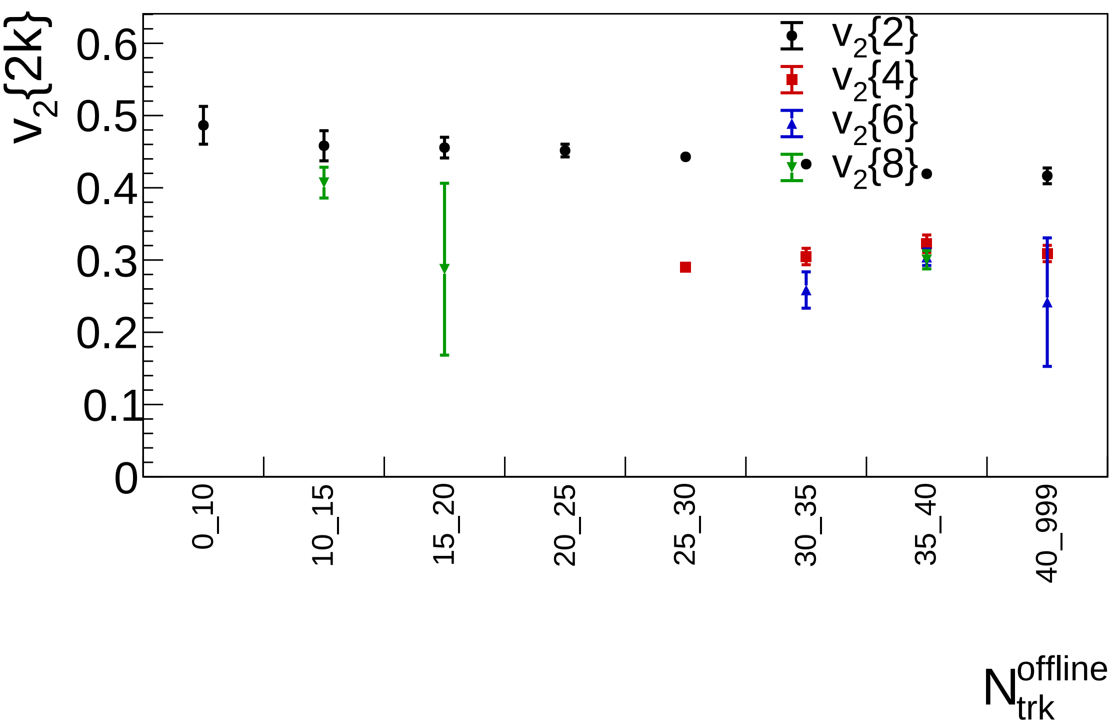



Figure 4: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ in the requested LEP2 merged sample using azimuth around the event thrust axis.

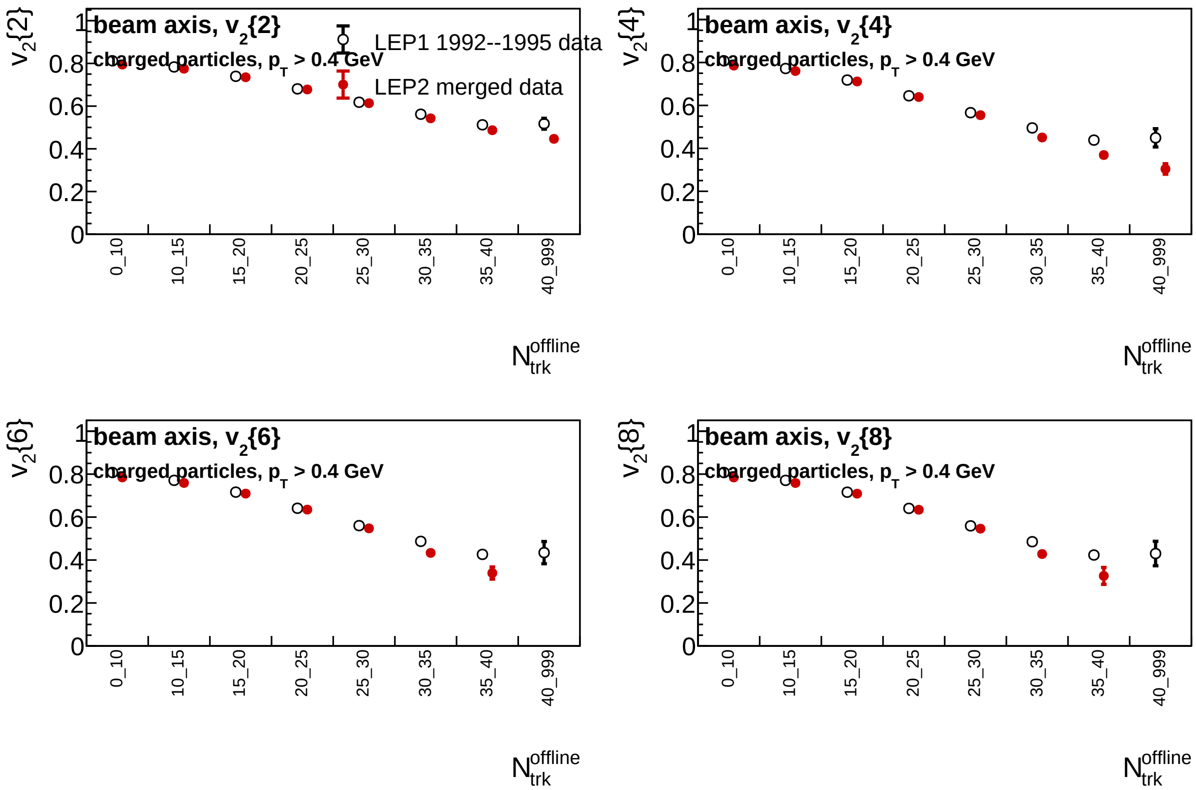



Figure 5: Detector-level beam-axis comparison of charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ between the merged LEP1 1992–1995 data sample and the requested LEP2 merged data sample. LEP1 is drawn as open circles and LEP2 as filled circles.

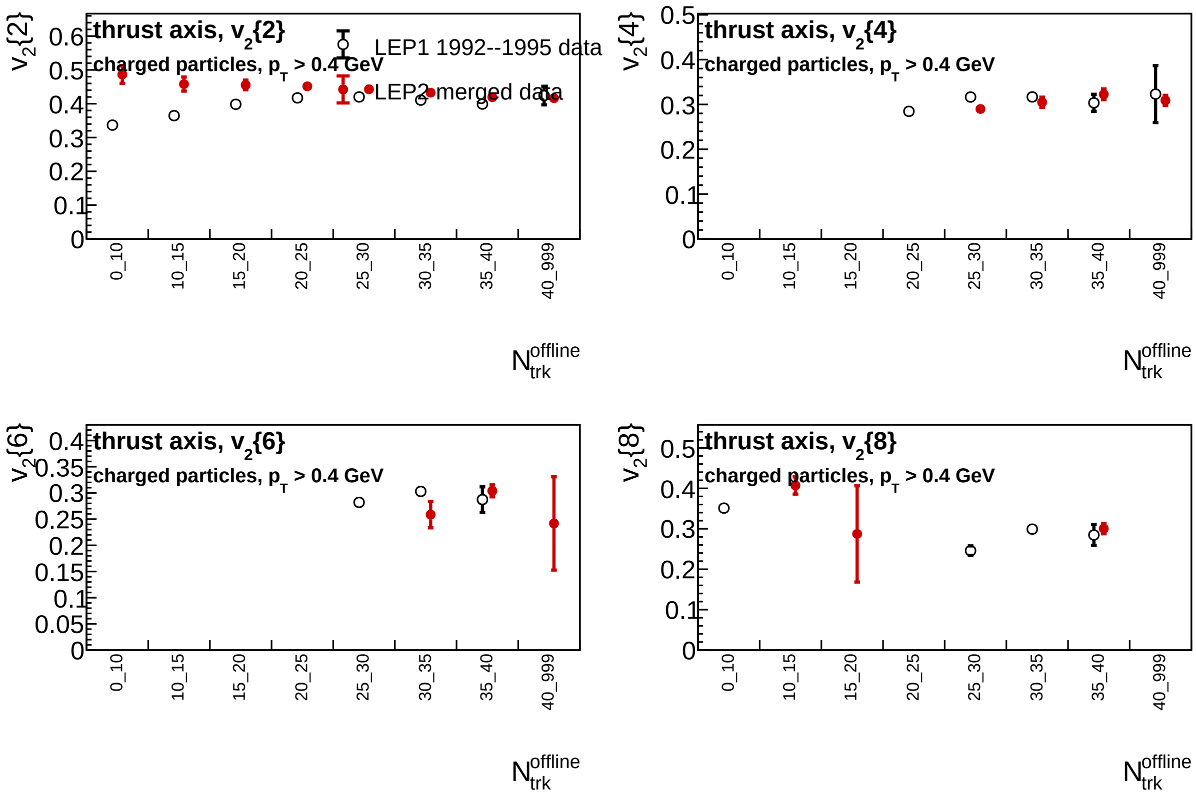



Figure 6: Detector-level thrust-axis comparison of charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ between the merged LEP1 1992–1995 data sample and the requested LEP2 merged data sample. LEP1 is drawn as open circles and LEP2 as filled circles.

This MC file contains 771,597 reconstructed entries in tree `t`, all from 1994. It is therefore used as a detector-level baseline rather than a strict year-matched comparison to the 1992–1995 data sample. After the charged-particle selection and multiplicity-bin requirement, 749,082 MC events enter the inclusive cumulant histograms.

Figures 7 and 8 show the standalone MC result for the beam and thrust axes. The beam-axis MC follows the same broad multiplicity dependence as the data-only result: the cumulant hierarchy is tight at low multiplicity and decreases with increasing multiplicity. In the thrust axis, the two-particle coefficient rises from low to intermediate multiplicity, while several low-multiplicity higher-order coefficients are sign-invalid and therefore not plotted.

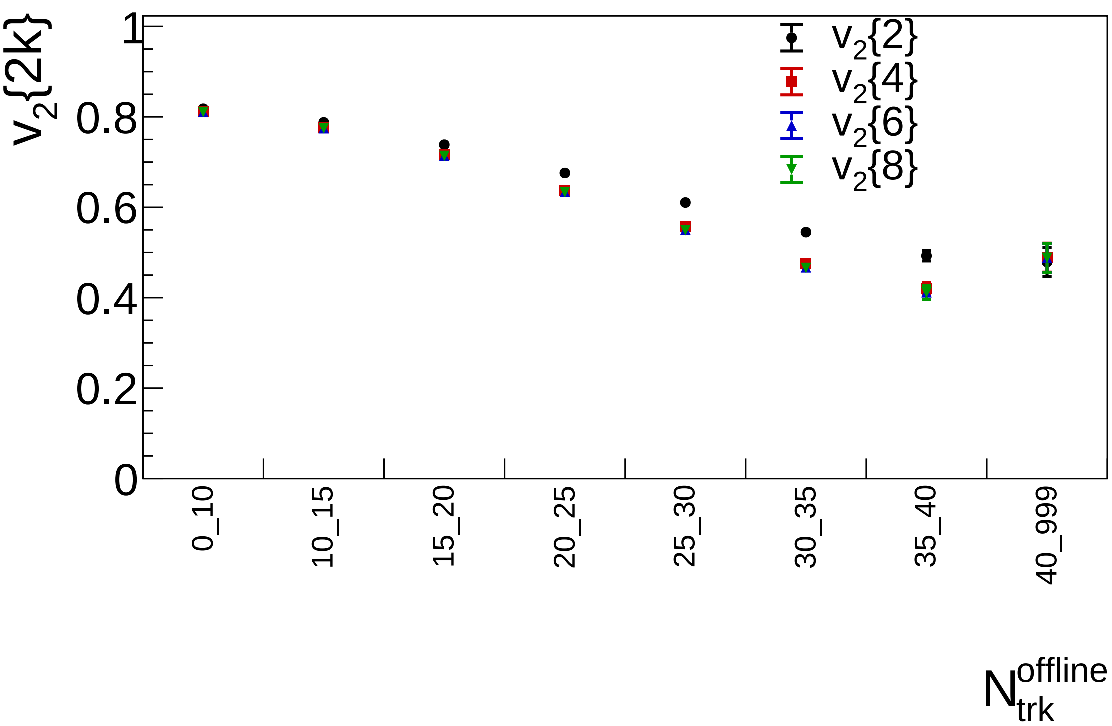



Figure 7: LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using the laboratory beam-axis azimuth.

Figures 9 and 10 compare the LEP1 1992–1995 data and 1994 reconstructed MC directly. For the beam axis, the data/MC agreement is at the percent level over most of the populated multiplicity range; the largest deviations appear in the highest-multiplicity bin, where the statistical uncertainty is also largest. For the thrust axis, $v_2\{2\}$ is well reproduced through intermediate multiplicity. The higher-order thrust-axis coefficients are more sensitive to cumulant sign changes and limited statistics, so missing points indicate bins where either data or MC did not yield a real value under the standard cumulant sign convention.

### 6.3 LEP1 MC nonflow-suppression closure

The reconstructed-MC comparison above shows that the nominal cumulants contain a large non-collective component. To test whether standard nonflow-suppression handles can make the LEP1 MC close to zero, the 1994 reconstructed-MC sample was reprocessed with a set of selections motivated by small-system flow analyses: large two-particle rapidity gaps, two-subevent cumulants, soft-track selections, charge-sign selections, and anti-jetty event-shape cuts. The rapidity-gap and subevent scans follow the logic of long-range two-particle and subevent-cumulant methods used in small collision systems [6, 7, 8].

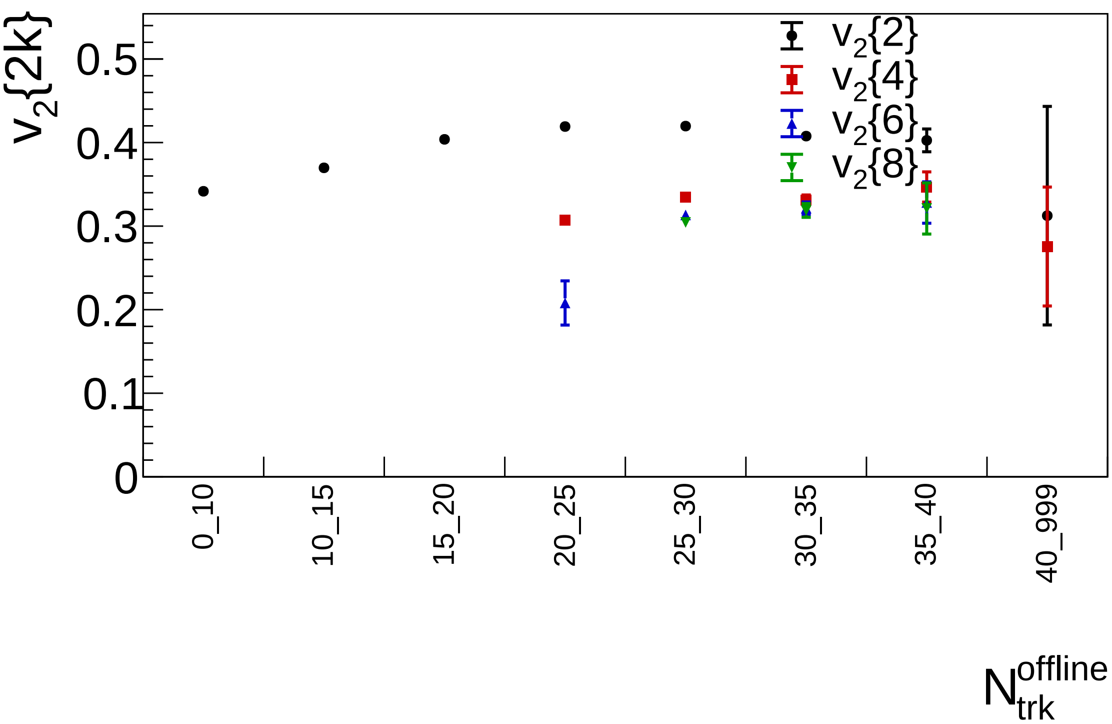



Figure 8: LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using azimuth around the event thrust axis.

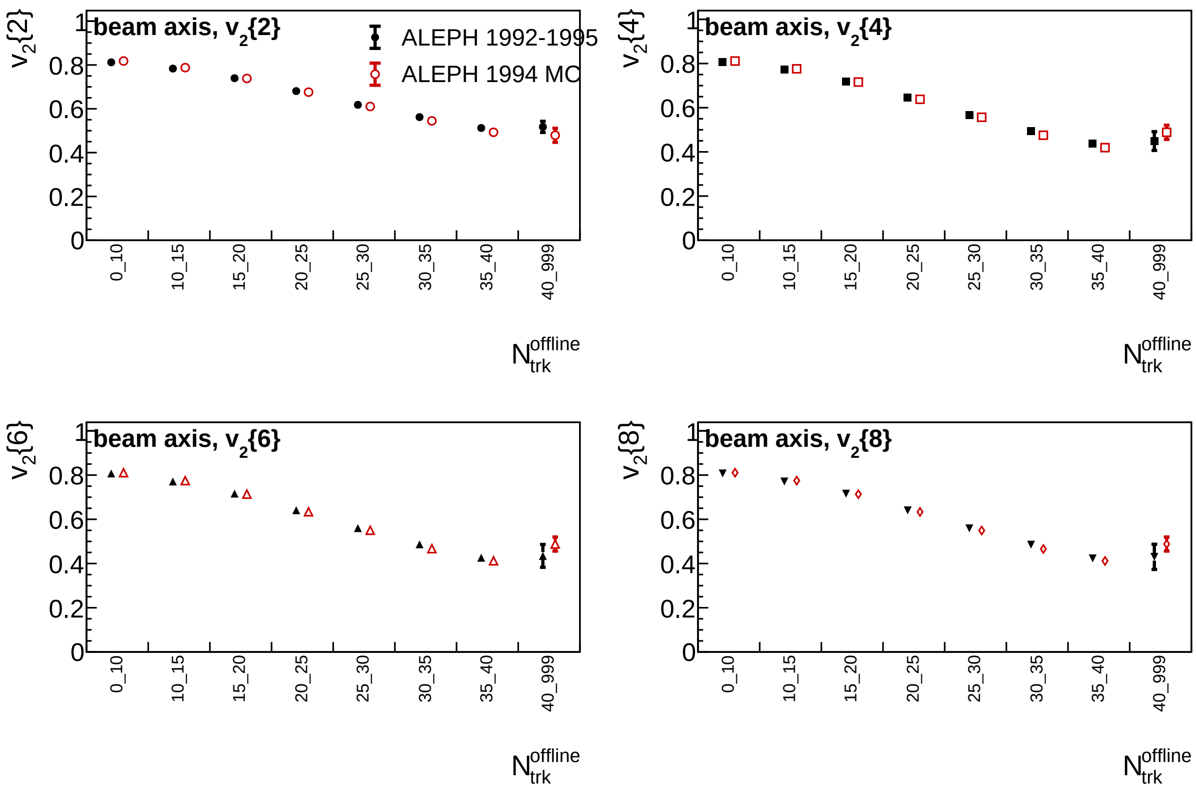



Figure 9: Beam-axis comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Each panel shows one cumulant order, with data and MC offset slightly in the lab-selected charged-multiplicity bin for readability.

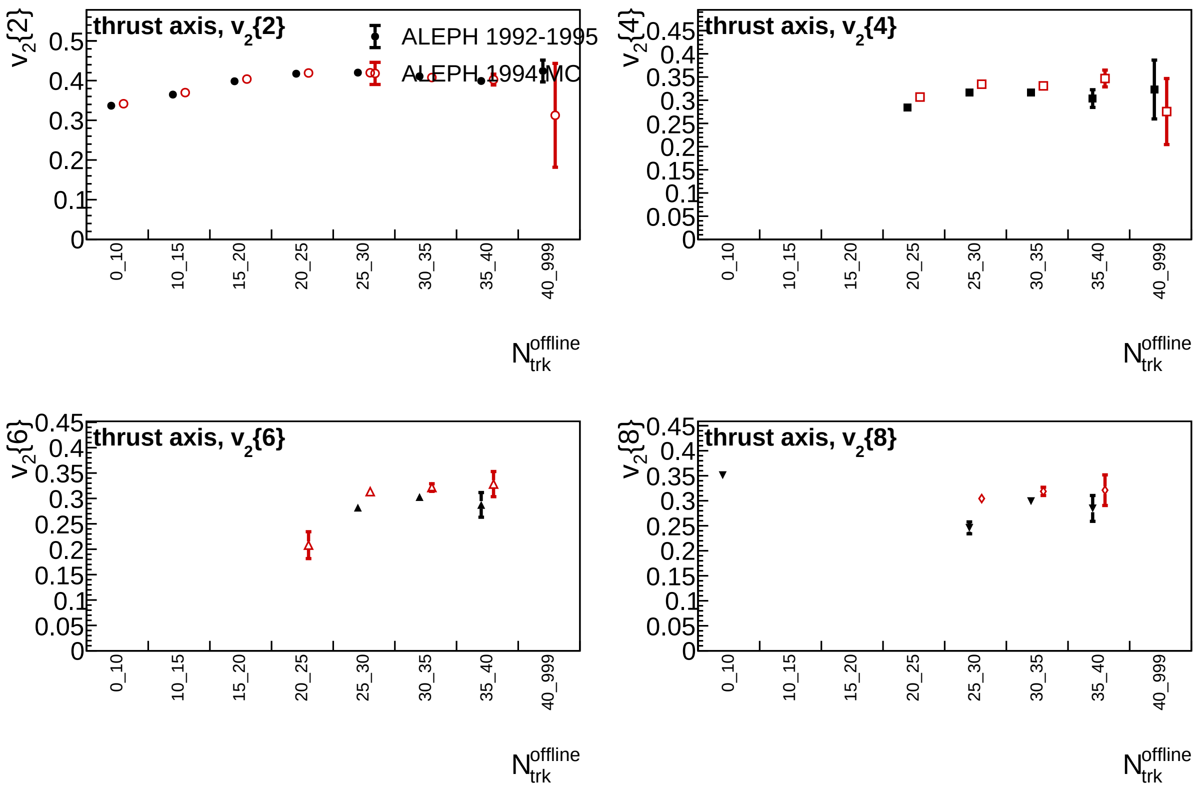



Figure 10: Thrust-axis comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Missing high-order points correspond to sign-invalid cumulants or bins without a valid real-valued $v_2\{m\}$.

The closure test is performed on the signed two-particle cumulant or signed rapidity-gap correlator, not only on $v_2\{2\}$. This is necessary because $v_2\{2\} = \sqrt{c_2\{2\}}$ is positive definite whenever $c_2\{2\} > 0$, so a positive $v_2\{2\}$ alone cannot establish whether the underlying MC correlator is compatible with zero. The first-pass MC production used the same multiplicity bins and sixteen-chunk jackknife procedure as the nominal MC comparison. For positive two-subevent boundary $b$, the two subevents are $\eta < -b$ and $\eta > b$, so the central excluded region corresponds to a subevent separation of $2b$.

The best-performing methods in this scan are summarized in Table 4. The strongest reduction is obtained from the largest rapidity-gap requirement, $|\Delta\eta| > 3.0$, especially when the azimuth and pseudorapidity are defined around the event thrust axis. However, none of the tested methods makes the signed MC correlator statistically consistent with zero in all multiplicity bins. Simple harmonic-level low-multiplicity and $1/N$-scaling subtractions were also tested as diagnostics; the fitted nonzero intercepts, $0.4550 \pm 0.0006$ for the beam axis and $0.1884 \pm 0.0006$ for the thrust axis, show that an integrated harmonic subtraction is not sufficient.

Table 4: Best first-pass LEP1 reconstructed-MC nonflow-suppression closure results. The ranking uses the RMS of the signed $c_2\{2\}$ or signed rapidity-gap correlator over valid multiplicity bins. A zero-compatible bin is one with absolute signed-correlator significance below two.

| Axis | Best tested method | RMS signed observable | Mean $v_2\{2\}$ | Zero-compatible bins |
|---|---|---|---|---|
| Beam | $\lvert\Delta\eta\rvert > 3.0$ | 0.128 | 0.383 | 2/7 |
| Beam | $0.4 < p_\mathrm{T} < 1$ GeV | 0.171 | 0.420 | 1/5 |
| Thrust | $\lvert\Delta\eta\rvert > 3.0$ | 0.0359 | 0.180 | 1/8 |
| Thrust | $0.4 < p_\mathrm{T} < 2$ GeV | 0.137 | 0.368 | 0/7 |

Figures 11 and 12 show the signed closure observables for the gap/subevent and track/event-shape families. The large-gap thrust-axis result is the closest to zero among the methods tested so far, but several multiplicity bins remain significantly positive. The next required closure checks are a genuine three-subevent $c_2\{4\}$ implementation and full $\Delta\phi$ template or $\Delta\eta$-binned correlation outputs; the current integrated cumulant outputs cannot reproduce the published template-subtraction procedures exactly.

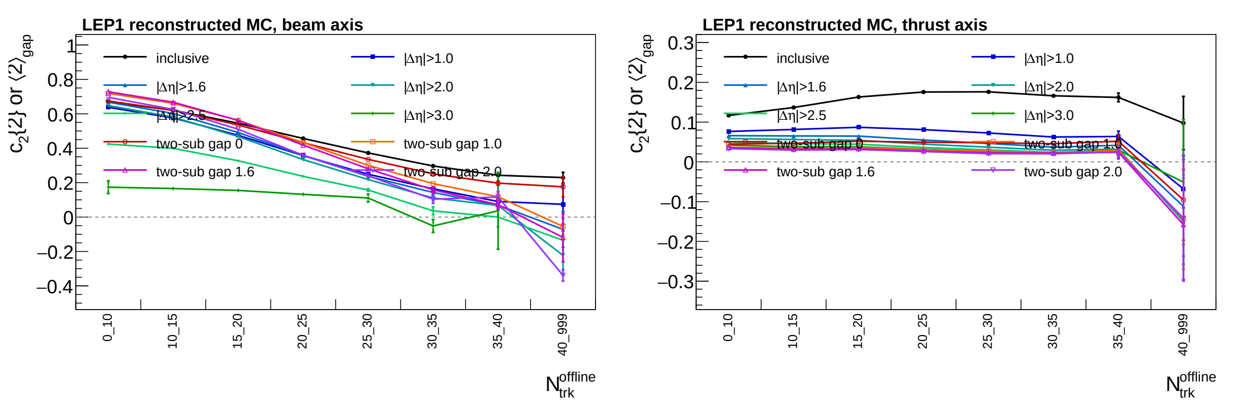



Figure 11: LEP1 reconstructed-MC signed two-particle closure observables for rapidity-gap and two-subevent nonflow-suppression scans. Left: beam axis. Right: thrust axis. The dashed line marks zero.

## 6.4 Generator-level Pythia shoving comparison

To isolate whether the string-shoving model can modify the cumulant observables, two generator-level Pythia 8.306 samples were produced from the local Gleipnir shoving branch. The current production uses five million accepted hadronic $e^+e^- \to \gamma^*/Z \to q\bar{q}$ events at $\sqrt{s} = 91.1876$ GeV per configuration. Each five-million-event sample is the merge of the original seed-12345 three-million-event production and an independent seed-67890 two-million-event extension. The merged ROOT trees were required to contain exactly

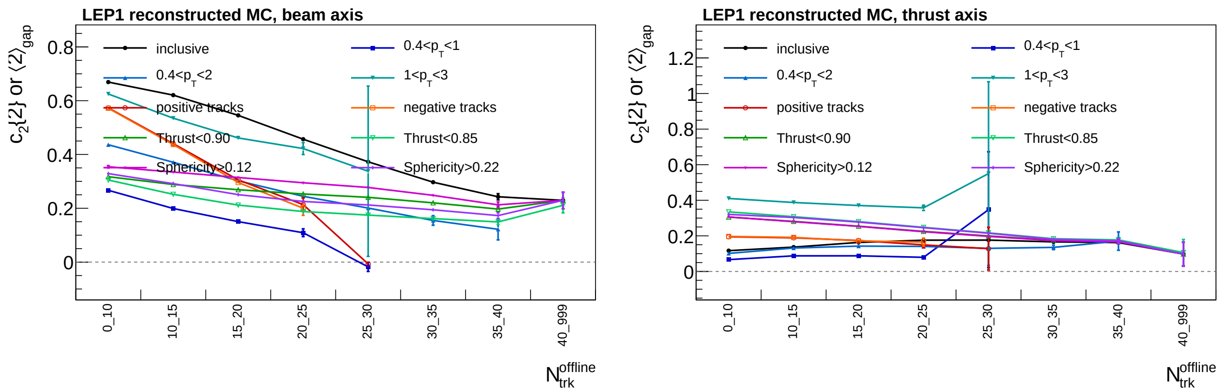



Figure 12: LEP1 reconstructed-MC signed two-particle closure observables for track-selection and event-shape nonflow-suppression scans. Left: beam axis. Right: thrust axis. The dashed line marks zero.

5,000,000 entries before the cumulant processing was run. Neutrinos are removed from the generator-level tree; charged particles are selected with the StudyMult-style $pwflag = 0, 1, 2$ convention. The cumulant selection, multiplicity bins, and sixteen-chunk jackknife procedure are identical for the no-shoving and shoving configurations.

The shoving sample enables the following settings:

- `StringInteractions:model = 3`
- `Gleipnir:shoving = on`
- `Gleipnir:StringR = 1.0`
- `Gleipnir:tauString = 1.0`
- `Gleipnir:tauHad = 2.0`
- `Gleipnir:nPushMaxCalc = 0`
- `Gleipnir:pushPT = 0.02`
- `Gleipnir:extendGluonRegions = on`
- `Gleipnir:repulsionFactor = 0.25`
- `Fragmentation:setVertices = on`
- `PartonVertex:setVertex = on`
- `PartonVertex:modeRadiation = 2`

The no-shoving control keeps the same hard process and shower settings but disables the Gleipnir block. Because turning shoving on changes the random-number consumption during hadronization, the comparison is treated as a matched-configuration comparison rather than an event-by-event paired subtraction.

The nominal $p_{\mathrm{T}} > 0.4$ GeV shoving effect is summarized in Table 5. The beam-axis cumulants change at the few-per-mille level with high statistical precision, while the thrust-axis cumulants show a larger increase. The quoted significance is (shoving $-$ no shoving)$/\sqrt{\sigma_{\mathrm{no}}^2 + \sigma_{\mathrm{sh}}^2}$, using the exported jackknife uncertainties and neglecting any covariance between the two generator configurations.

## 6.5 Generator-level Pythia shoving nonflow-suppression checks

The same five-million-event Pythia samples were processed with the nonflow-suppression handles used for the LEP1 MC closure study. The two-particle rapidity-gap scans use $|\Delta\eta| > 1.0, 1.6, 2.0, 2.5, 3.0$. The two-subevent cumulants split each axis frame into $\eta < -b$ and $\eta > b$, with $b = 0, 0.5, 0.8, 1.0$, corresponding to no excluded middle region and total gaps of 1.0, 1.6, and 2.0. Track-selection scans use $0.4 < p_{\mathrm{T}} < 1$ GeV, $0.4 < p_{\mathrm{T}} < 2$ GeV, $1 < p_{\mathrm{T}} < 3$ GeV, positive charges only, and negative charges only. Event-shape scans use

Table 5: Largest observed generator-level Pythia shoving effect in the current five-million-event nominal comparison. Only bins with real, positive plotted $v_2\{m\}$ values are used.

| Axis | Observable | Most sensitive bin | Shoving/no-shoving | Max. significance |
|---|---|---|---|---|
| Beam | $v_2\{2\}$ | 20–25 | 1.0047 | 11.55 |
| Beam | $v_2\{4\}$ | 10–15 | 1.0030 | 8.99 |
| Beam | $v_2\{6\}$ | 10–15 | 1.0031 | 8.96 |
| Beam | $v_2\{8\}$ | 10–15 | 1.0031 | 8.96 |
| Thrust | $v_2\{2\}$ | 15–20 | 1.0161 | 21.40 |
| Thrust | $v_2\{4\}$ | 25–30 | 1.0315 | 5.04 |
| Thrust | $v_2\{6\}$ | 25–30 | 1.0525 | 4.24 |
| Thrust | $v_2\{8\}$ | 25–30 | 1.0852 | 3.58 |

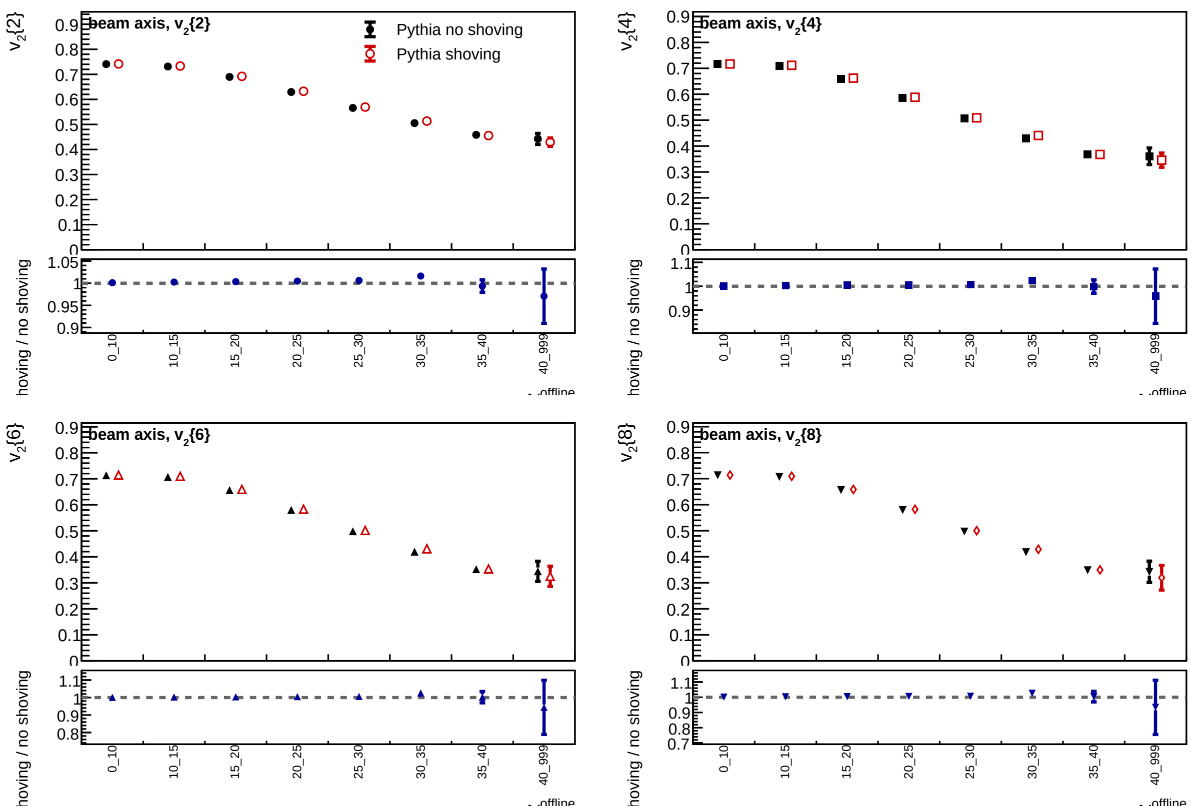



Figure 13: Generator-level Pythia comparison of beam-axis $v_2\{m\}$ with and without Gleipnir shoving. Both samples contain five million accepted hadronic Z-pole events and are processed with the same charged-particle selection and jackknife chunking. The lower panel in each observable pad gives the shoving/no-shoving ratio.

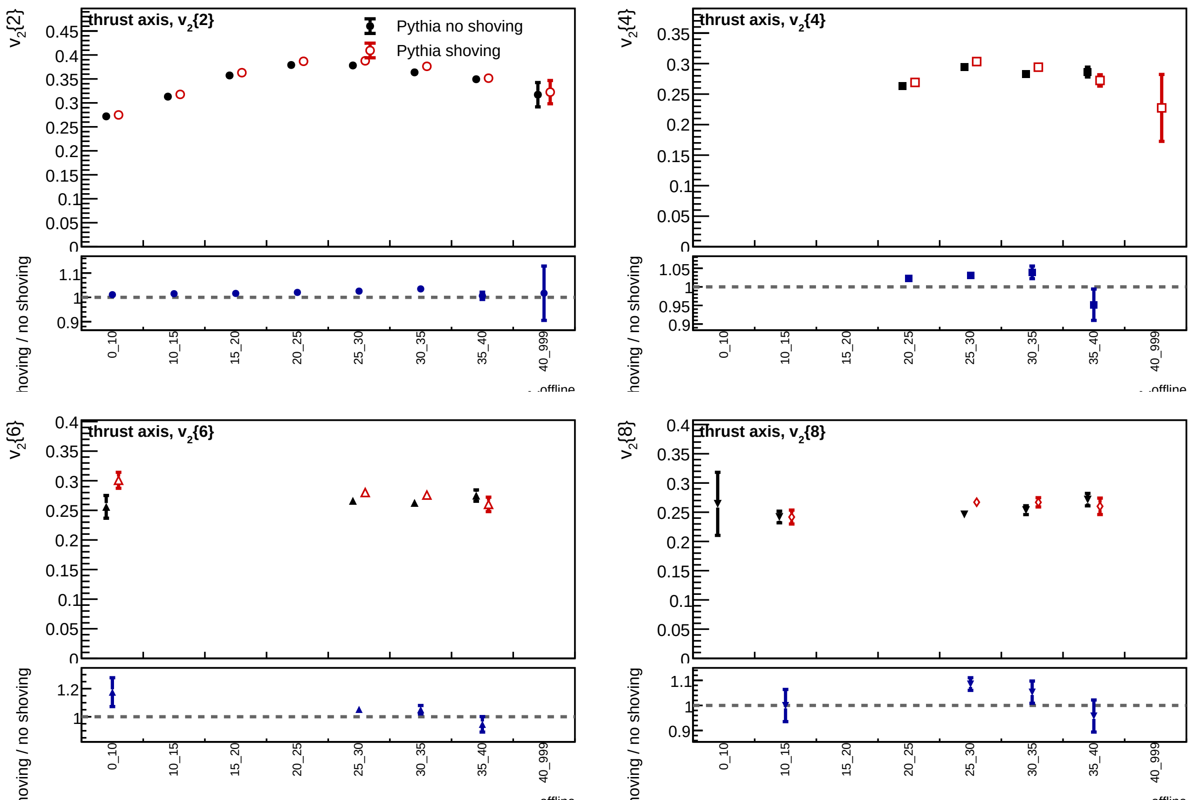



Figure 14: Generator-level Pythia comparison of thrust-axis $v_2\{m\}$ with and without Gleipnir shoving. The shoving sample gives a visible increase in the thrust-axis cumulants, especially for $v_2\{2\}$ at intermediate multiplicity. The lower panel in each observable pad gives the shoving/no-shoving ratio.

$T < 0.90$, $T < 0.85$, $S > 0.12$, and $S > 0.22$. For the generated vector trees, the analyzer computes thrust and sphericity from visible final-particle momenta when those branches are not present.

Table 6 gives the most significant $v_2\{2\}$-level shoving/no-shoving difference for each suppression handle and axis. For rapidity-gap rows the observable is the gap $v_2\{2\}$; for two-subevent rows it is the two-subevent $v_2\{2\}$; otherwise it is the full-event $v_2\{2\}$. The shoving enhancement is not removed by soft-track, charge, or event-shape selections. Large rapidity gaps and the two-subevent construction reduce the thrust-axis significance, with the strongest tested suppression coming from the two-subevent gap of 2.0, where the thrust-axis $v_2\{2\}$ difference is reduced to about 1.7 standard deviations. These checks show that the shoving/no-shoving difference survives many standard nonflow handles, but they do not demonstrate that the absolute Pythia cumulants are collective; the no-shoving generator still contains sizeable jet/topology correlations. All comparison figures in this subsection include the shoving/no-shoving ratio. For the rapidity-gap figures, the middle panel keeps the gap/inclusive diagnostic and the bottom panel shows the shoving/no-shoving ratio for the inclusive and gap observables.

Table 6: Five-million-event Pythia shoving/no-shoving comparison under nonflow-suppression handles. Entries are shoving/no-shoving ratios for the most significant $v_2\{2\}$-level bin, with the significance in parentheses.

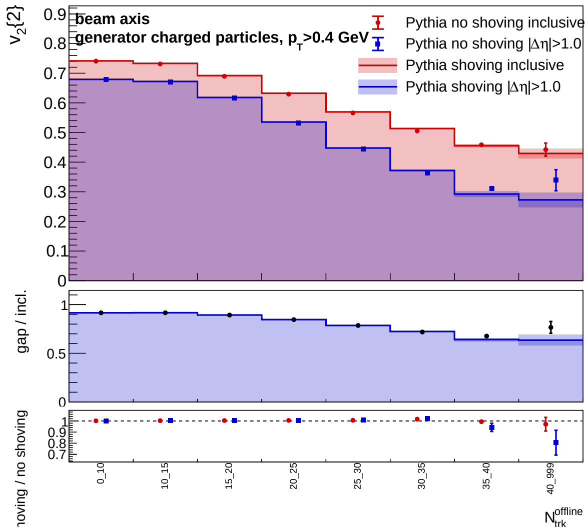
| Technique | Case | Beam axis | Thrust axis |
|---|---|---|---|
| Nominal | $p_T > 0.4$ GeV | 1.0047 (11.55) | 1.0161 (21.40) |
| Rapidity gap | $\|\Delta\eta\| > 1.0$ | 1.0039 (6.57) | 1.0195 (11.60) |
| Rapidity gap | $\|\Delta\eta\| > 1.6$ | 1.0032 (5.96) | 1.0195 (6.67) |
| Rapidity gap | $\|\Delta\eta\| > 2.0$ | 1.0037 (6.17) | 1.0219 (5.52) |
| Rapidity gap | $\|\Delta\eta\| > 2.5$ | 1.0037 (4.31) | 1.0215 (3.38) |
| Rapidity gap | $\|\Delta\eta\| > 3.0$ | 1.0033 (2.75) | 1.0272 (2.73) |
| Two subevent | $\eta < 0$ vs. $\eta > 0$ | 1.0047 (9.25) | 1.0307 (6.35) |
| Two subevent | gap 1.0 | 1.0025 (6.39) | 1.0554 (3.76) |
| Two subevent | gap 1.6 | 1.0026 (5.78) | 1.0563 (2.60) |
| Two subevent | gap 2.0 | 1.0026 (4.61) | 1.0364 (1.73) |
| Track selection | $0.4 < p_T < 1$ GeV | 1.0090 (16.13) | 1.0319 (30.28) |
| Track selection | $0.4 < p_T < 2$ GeV | 1.0068 (12.27) | 1.0219 (22.30) |
| Track selection | $1 < p_T < 3$ GeV | 1.0028 (9.85) | 1.0080 (13.67) |
| Charge selection | positive | 1.0052 (11.15) | 1.0188 (16.35) |
| Charge selection | negative | 1.0031 (11.38) | 1.0196 (17.57) |
| Event shape | $T < 0.90$ | 1.0044 (5.40) | 1.0194 (14.59) |
| Event shape | $T < 0.85$ | 1.0071 (4.74) | 1.0191 (14.51) |
| Event shape | $S > 0.12$ | 1.0220 (5.59) | 1.0204 (13.51) |
| Event shape | $S > 0.22$ | 1.0087 (5.82) | 1.0189 (10.97) |

## 6.6 Two-particle rapidity-gap comparison

The inclusive two-particle result contains every ordered charged-particle pair in an event. To test the sensitivity to short-range and same-jet correlations, the LEP1 1992–1995 data and 1994 reconstructed-MC samples were reprocessed with an additional two-particle rapidity-gap observable. The track selection, axis definitions, charged-multiplicity bins, and sixteen event chunks are the same as in the inclusive data/MC comparison. The default additional requirement is a strict pair cut

$$|\eta_i - \eta_j| > 2.0. \tag{12}$$

A second production is also run with $|\eta_i - \eta_j| > 1.6$ to test the stability of the rapidity-gap suppression against the gap threshold. In both cases, $\eta$ is evaluated in the same coordinate system as the azimuthal angle: laboratory pseudorapidity for the beam-axis result and pseudorapidity around the reconstructed thrust axis for the thrust-axis result. The horizontal axis remains the selected laboratory charged-particle multiplicity, matching the other figures in this note.



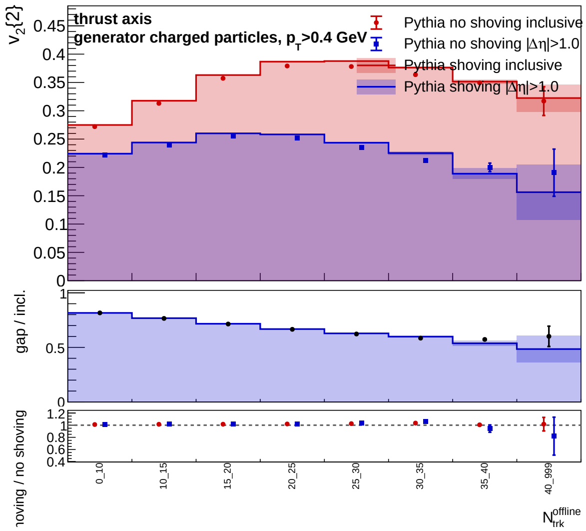
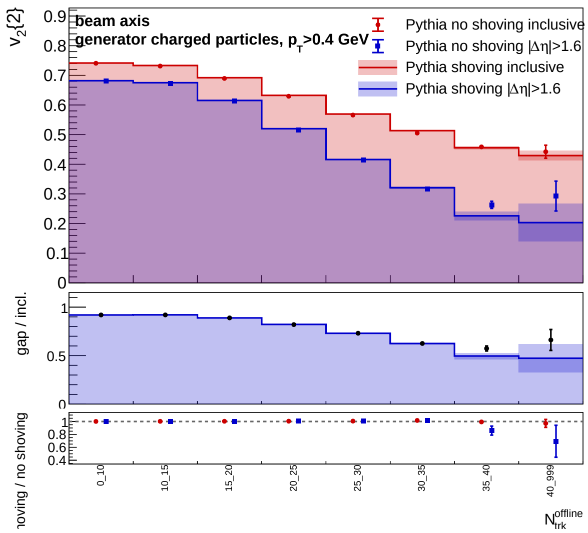




Figure 15: Generator-level Pythia no-shoving/shoving comparison for Two-particle rapidity-gap selection $|\Delta\eta| > 1.0$. Left: beam axis. Right: thrust axis.



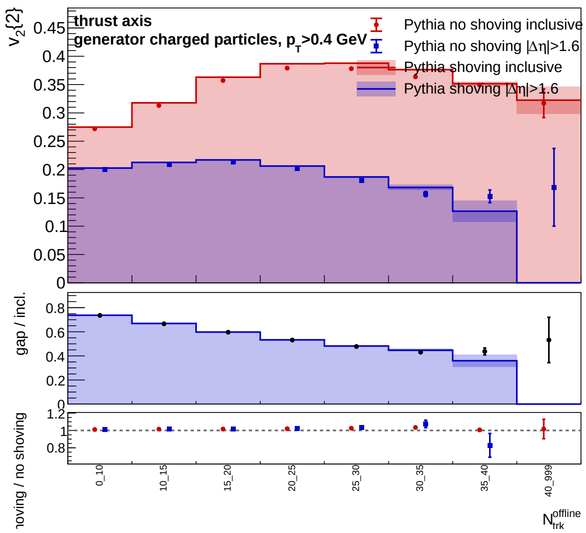




Figure 16: Generator-level Pythia no-shoving/shoving comparison for Two-particle rapidity-gap selection $|\Delta\eta| > 1.6$. Left: beam axis. Right: thrust axis.

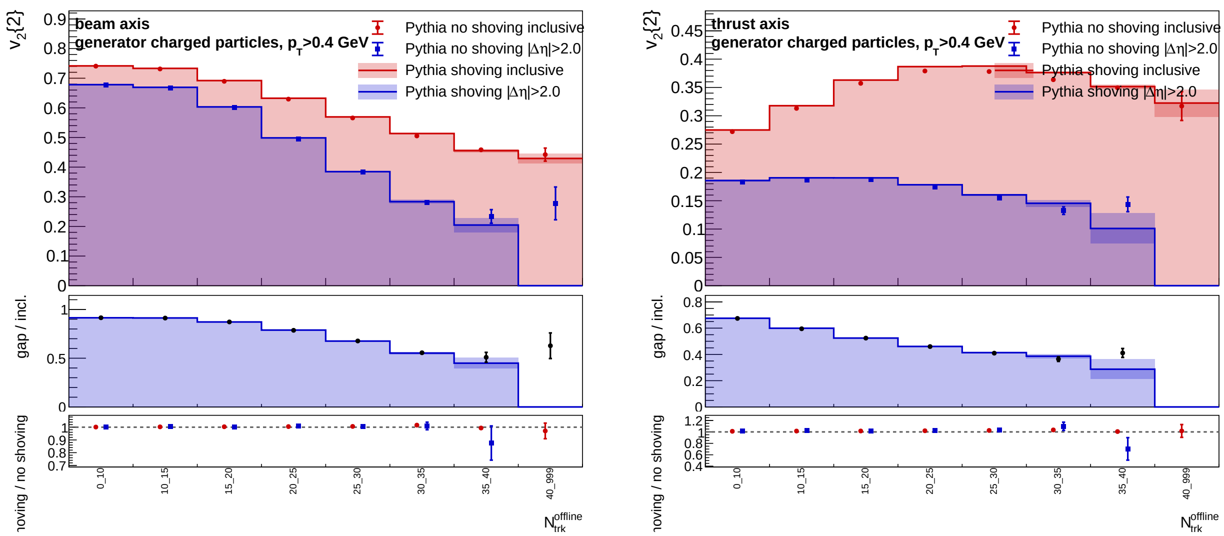



Figure 17: Generator-level Pythia no-shoving/shoving comparison for Two-particle rapidity-gap selection $|\Delta\eta| > 2.0$. Left: beam axis. Right: thrust axis.

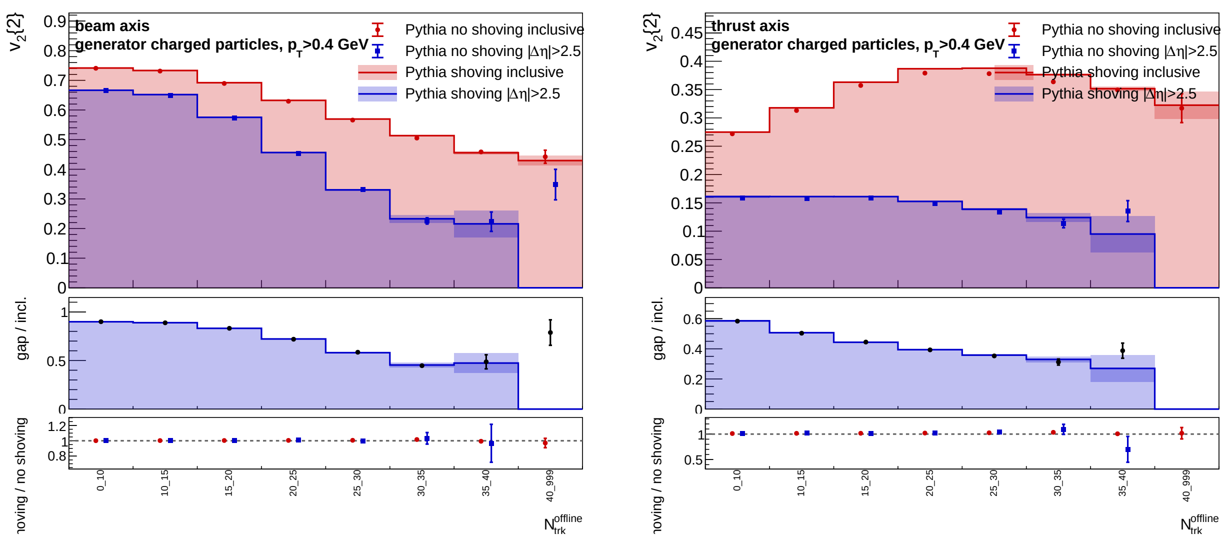



Figure 18: Generator-level Pythia no-shoving/shoving comparison for Two-particle rapidity-gap selection $|\Delta\eta| > 2.5$. Left: beam axis. Right: thrust axis.

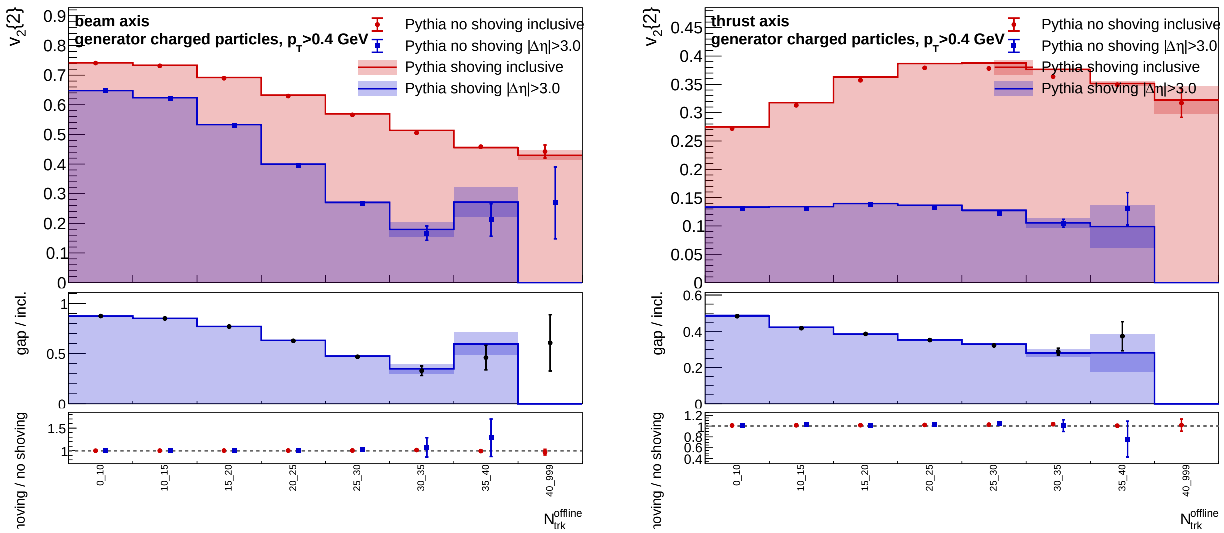



Figure 19: Generator-level Pythia no-shoving/shoving comparison for Two-particle rapidity-gap selection $|\Delta\eta| > 3.0$. Left: beam axis. Right: thrust axis.

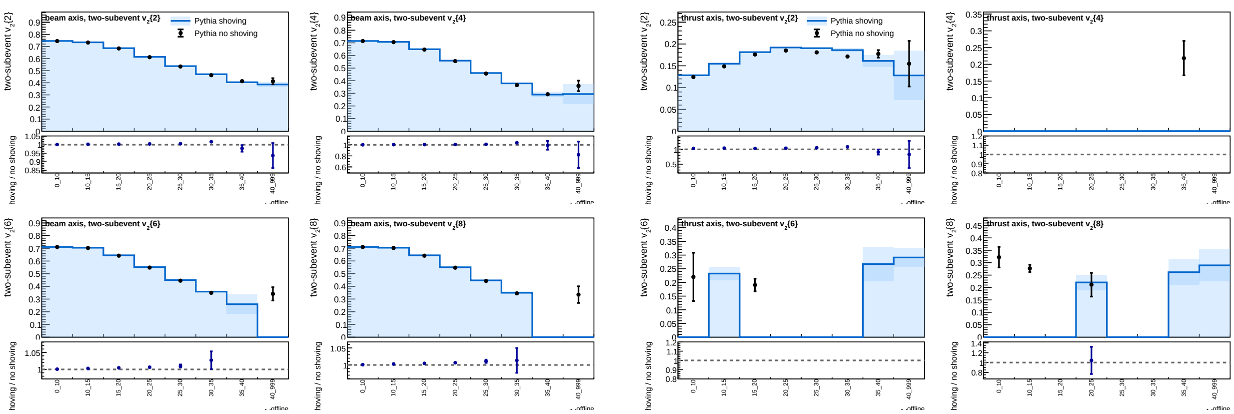



Figure 20: Generator-level Pythia no-shoving/shoving comparison for Two-subevent cumulants with $\eta < 0$ versus $\eta > 0$. Left: beam axis. Right: thrust axis.

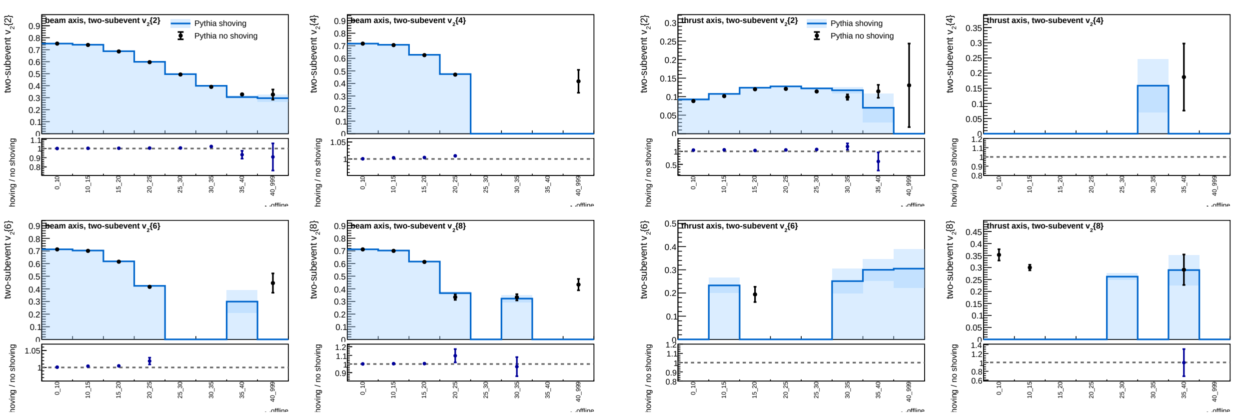



Figure 21: Generator-level Pythia no-shoving/shoving comparison for Two-subevent cumulants with total excluded middle-region gap 1.0. Left: beam axis. Right: thrust axis.

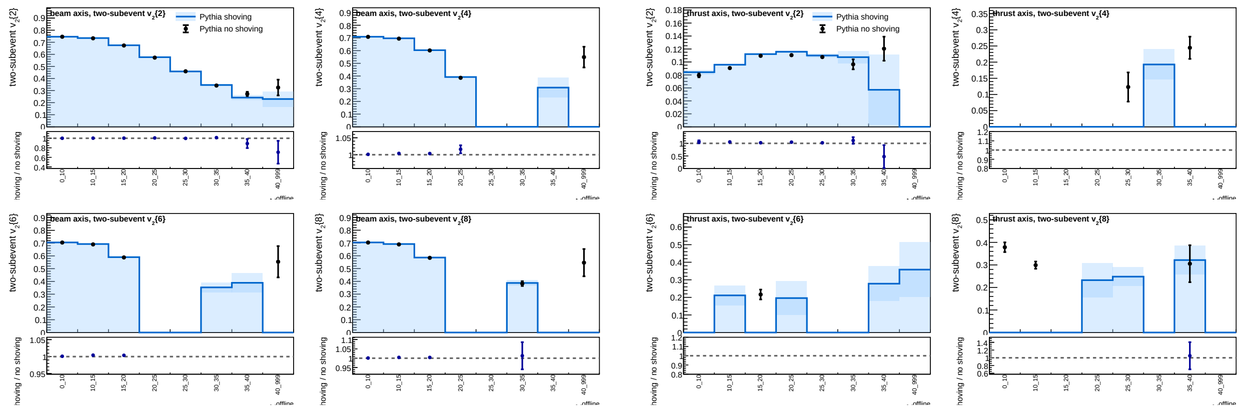


Figure 22: Generator-level Pythia no-shoving/shoving comparison for Two-subevent cumulants with total excluded middle-region gap 1.6. Left: beam axis. Right: thrust axis.

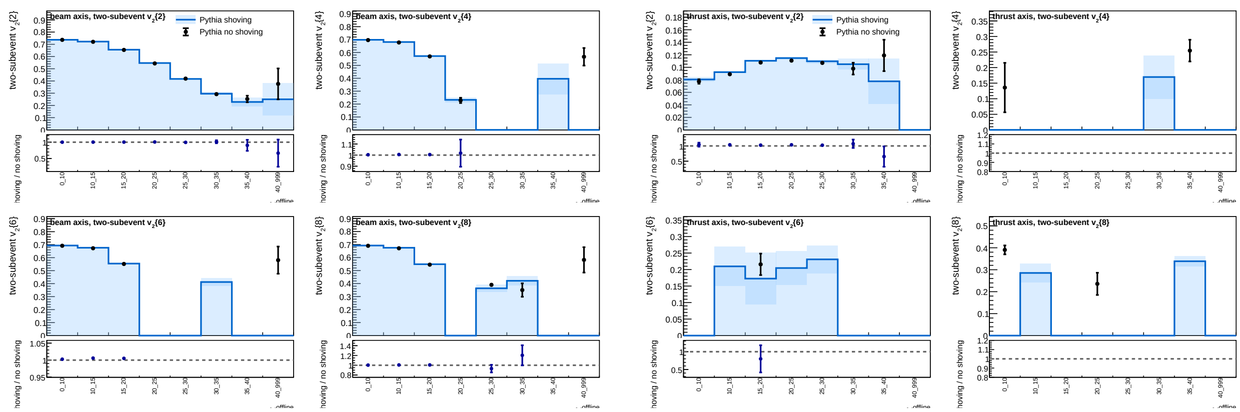


Figure 23: Generator-level Pythia no-shoving/shoving comparison for Two-subevent cumulants with total excluded middle-region gap 2.0. Left: beam axis. Right: thrust axis.

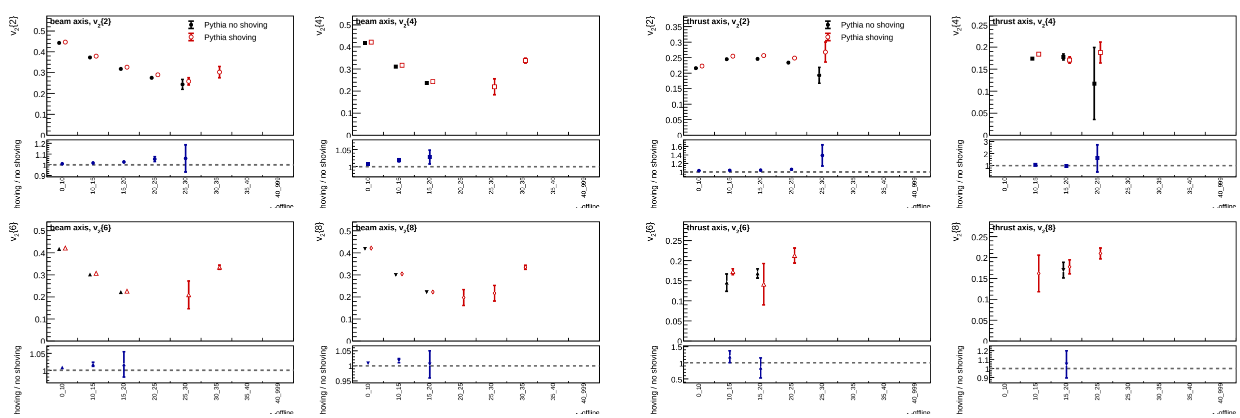


Figure 24: Generator-level Pythia no-shoving/shoving comparison for Track selection $0.4 < p_{\mathrm{T}} < 1$ GeV. Left: beam axis. Right: thrust axis.

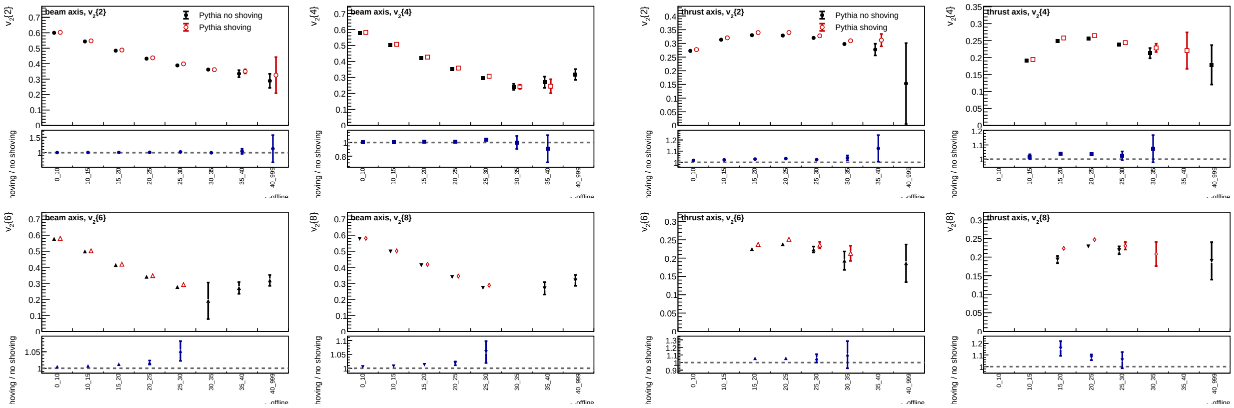



Figure 25: Generator-level Pythia no-shoving/shoving comparison for Track selection $0.4 < p_{\mathrm{T}} < 2$ GeV. Left: beam axis. Right: thrust axis.

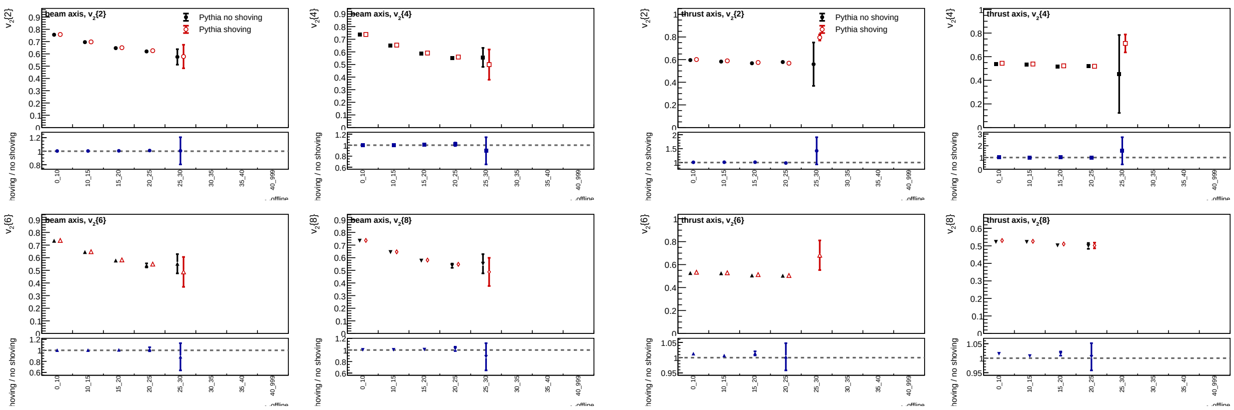



Figure 26: Generator-level Pythia no-shoving/shoving comparison for Track selection $1 < p_{\mathrm{T}} < 3$ GeV. Left: beam axis. Right: thrust axis.

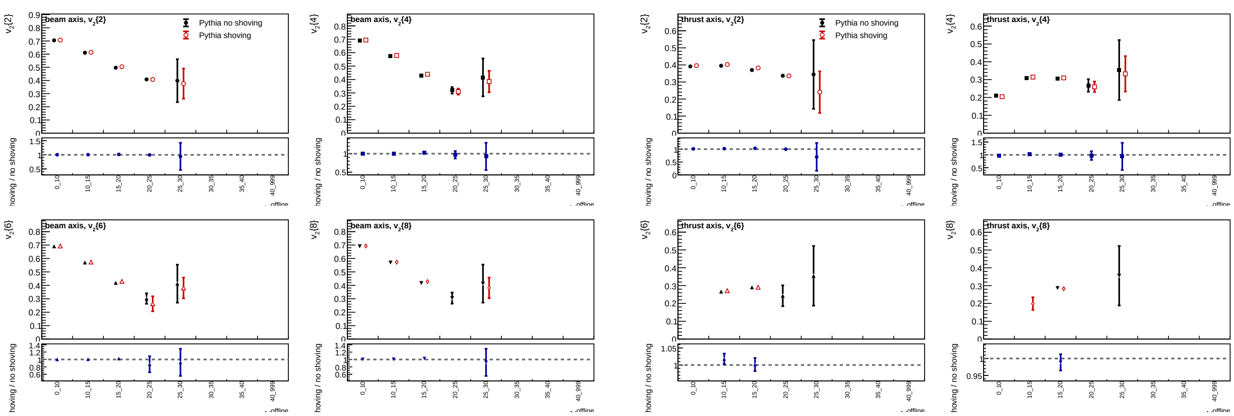



Figure 27: Generator-level Pythia no-shoving/shoving comparison for Positive charged particles only. Left: beam axis. Right: thrust axis.

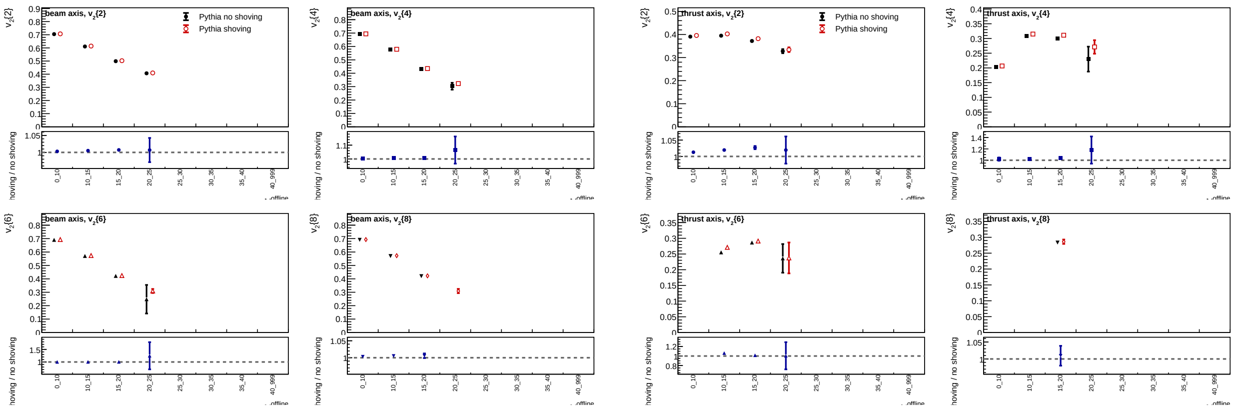



Figure 28: Generator-level Pythia no-shoving/shoving comparison for Negative charged particles only. Left: beam axis. Right: thrust axis.

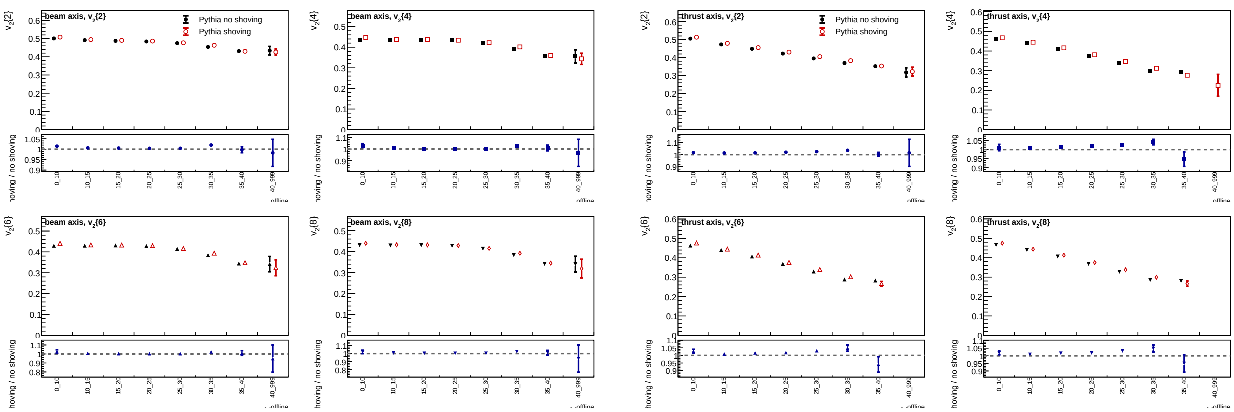



Figure 29: Generator-level Pythia no-shoving/shoving comparison for Event-shape selection $T < 0.90$. Left: beam axis. Right: thrust axis.

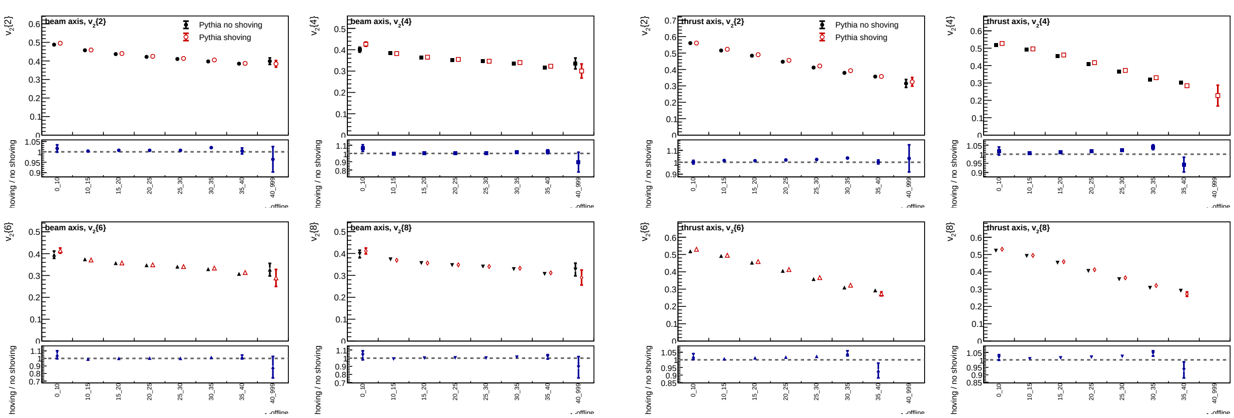



Figure 30: Generator-level Pythia no-shoving/shoving comparison for Event-shape selection $T < 0.85$. Left: beam axis. Right: thrust axis.

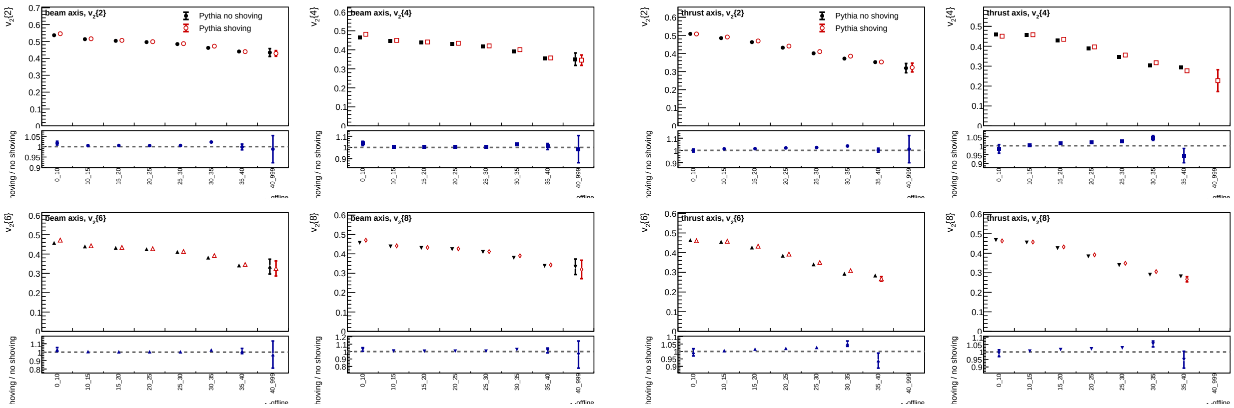



Figure 31: Generator-level Pythia no-shoving/shoving comparison for Event-shape selection $S > 0.12$. Left: beam axis. Right: thrust axis.

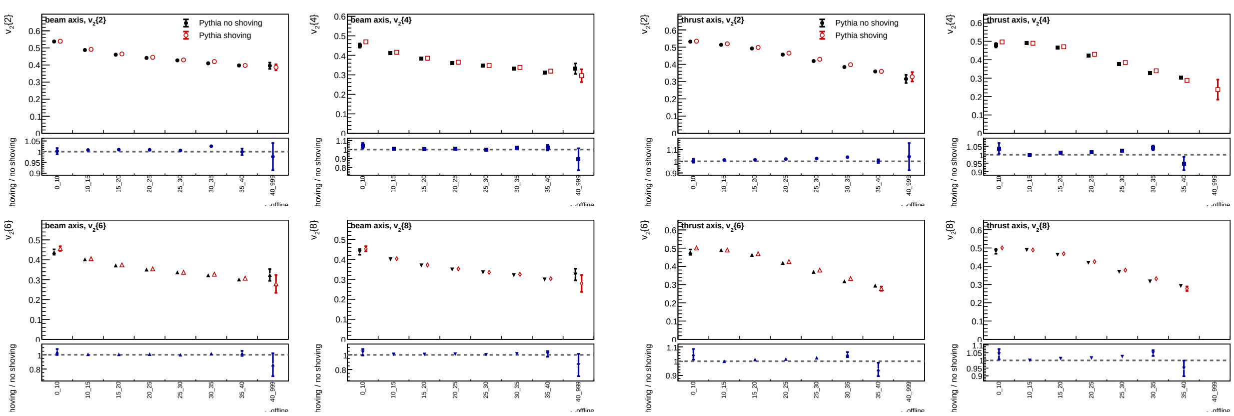



Figure 32: Generator-level Pythia no-shoving/shoving comparison for Event-shape selection $S > 0.22$. Left: beam axis. Right: thrust axis.

For every event and multiplicity bin, the analyzer stores the mergeable sums

$$N_2^{\Delta\eta>2} = \sum_{\text{events}} \sum_{i\neq j} \Theta(|\eta_i - \eta_j| - 2.0) \cos\left[2(\phi_i - \phi_j)\right], \tag{13}$$

$$D_2^{\Delta\eta>2} = \sum_{\text{events}} \sum_{i\neq j} \Theta(|\eta_i - \eta_j| - 2.0), \tag{14}$$

with ordered-pair weights. Equivalently, the implementation loops over unordered pairs and adds both ordered partners through a factor of two in the numerator and denominator. The gap correlation and flow coefficient are then computed only in the summary step,

$$\langle 2\rangle_{|\Delta\eta|>2} = \frac{N_2^{\Delta\eta>2}}{D_2^{\Delta\eta>2}}, \qquad v_2\{2, |\Delta\eta| > 2\} = \sqrt{\langle 2\rangle_{|\Delta\eta|>2}}, \tag{15}$$

when the correlation is non-negative. The inclusive $v_2\{2\}$ and the rapidity-gap $v_2\{2, |\Delta\eta| > 2\}$ are both recomputed for each leave-one-chunk jackknife sample, so the plotted uncertainties include the nonlinear square-root transformation. The lower-panel gap/inclusive ratio is also formed in each leave-one sample and jackknifed as a ratio, preserving the statistical covariance between the inclusive and rapidity-gap measurements.

Figures 33 and 34 compare the inclusive and rapidity-gap measurements. MC is drawn as histogram bands and data as points. In the beam axis, the rapidity-gap value is close to the inclusive value at low multiplicity and becomes smaller from roughly 15 charged tracks onward; the data gap/inclusive ratio falls to about 0.65 in the 30–35 bin, with the high-multiplicity tail limited by statistics. The reconstructed MC follows the same trend. In the thrust axis, the rapidity-gap suppression is stronger: the data gap/inclusive ratio decreases from about 0.71 in the 0–10 bin to about 0.43 around 30–35 charged tracks, again with consistent MC behavior within the quoted statistical uncertainties. This pattern is consistent with a sizable contribution to the inclusive two-particle coefficient from short-range or jet-like correlations that are reduced by the large $|\Delta\eta|$ separation.

Figures 35 and 36 show the corresponding comparison for $|\Delta\eta| > 1.6$. As expected, the looser gap admits more event pairs and therefore gives a less suppressed gap/inclusive ratio than the $|\Delta\eta| > 2$ selection, while preserving the same qualitative multiplicity dependence.

## 6.7 Track-pT-restricted cross-check

As a first charged-track kinematic variation, the inclusive and rapidity-gap productions were repeated with selected charged particles restricted to $1 < p_{\mathrm{T}} < 3$ GeV. The beam-axis selection uses the laboratory transverse momentum, while the thrust-axis selection uses transverse momentum relative to the reconstructed event thrust axis. The event samples, charged-multiplicity bins, and sixteen-chunk jackknife procedure are otherwise unchanged. The default rapidity-gap rerun uses $|\Delta\eta| > 2.0$, and a separate rerun uses $|\Delta\eta| > 1.6$.

Figures 37 and 38 show the resulting $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ comparisons. The tighter track-$p_{\mathrm{T}}$ window reduces the selected charged multiplicity, so the highest offline-multiplicity bins are sparsely populated and several higher-order cumulant points become sign-invalid or statistically unusable. In the populated region, the beam-axis $v_2\{2\}$ data/MC ratio remains close to unity, while the high-order and thrust-axis points should be interpreted as detector-level cross-checks with limited high-multiplicity reach.

Figures 39 and 40 compare inclusive $v_2\{2\}$ with the $|\Delta\eta| > 2.0$ rapidity-gap value in the same track-$p_{\mathrm{T}}$ window. Figures 41 and 42 show the corresponding $|\Delta\eta| > 1.6$ rerun. The looser gap gives a less suppressed gap/inclusive ratio, as in the nominal $p_{\mathrm{T}} > 0.4$ GeV selection. The uncertainties on the gap/inclusive and data/MC ratios are evaluated with the same leave-one-chunk jackknife ratio procedure used for the nominal result.

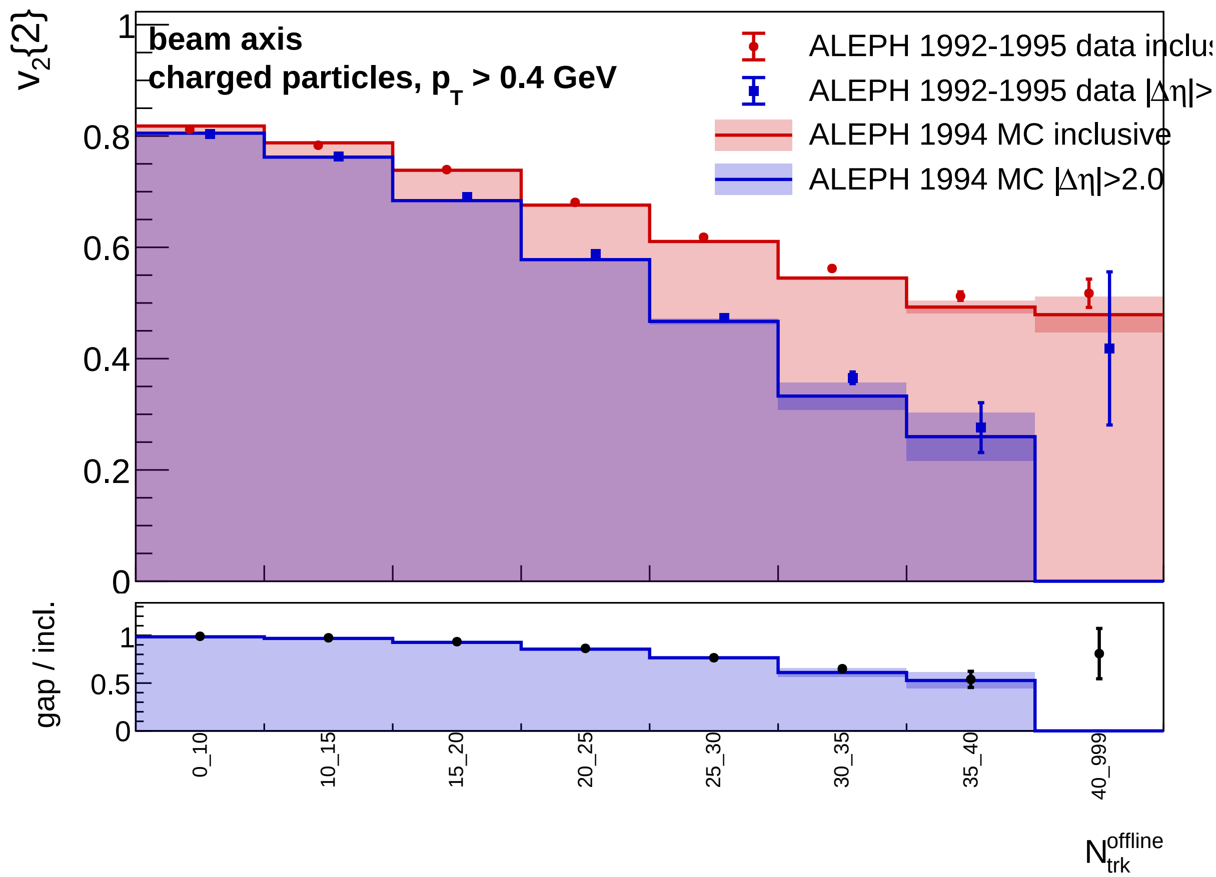



Figure 33: Beam-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 2\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. MC is shown as histogram bands and data as points. The lower panel shows the rapidity-gap value divided by the inclusive value.

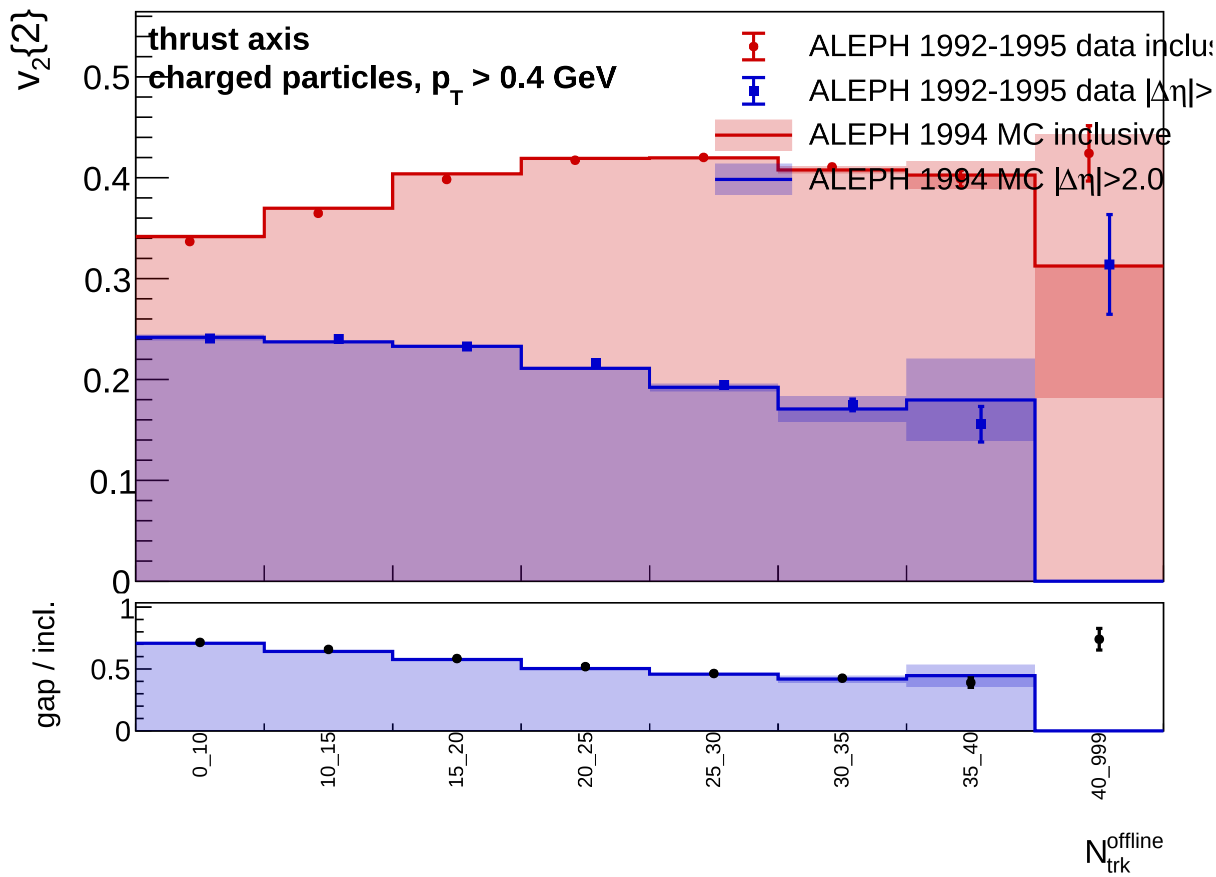
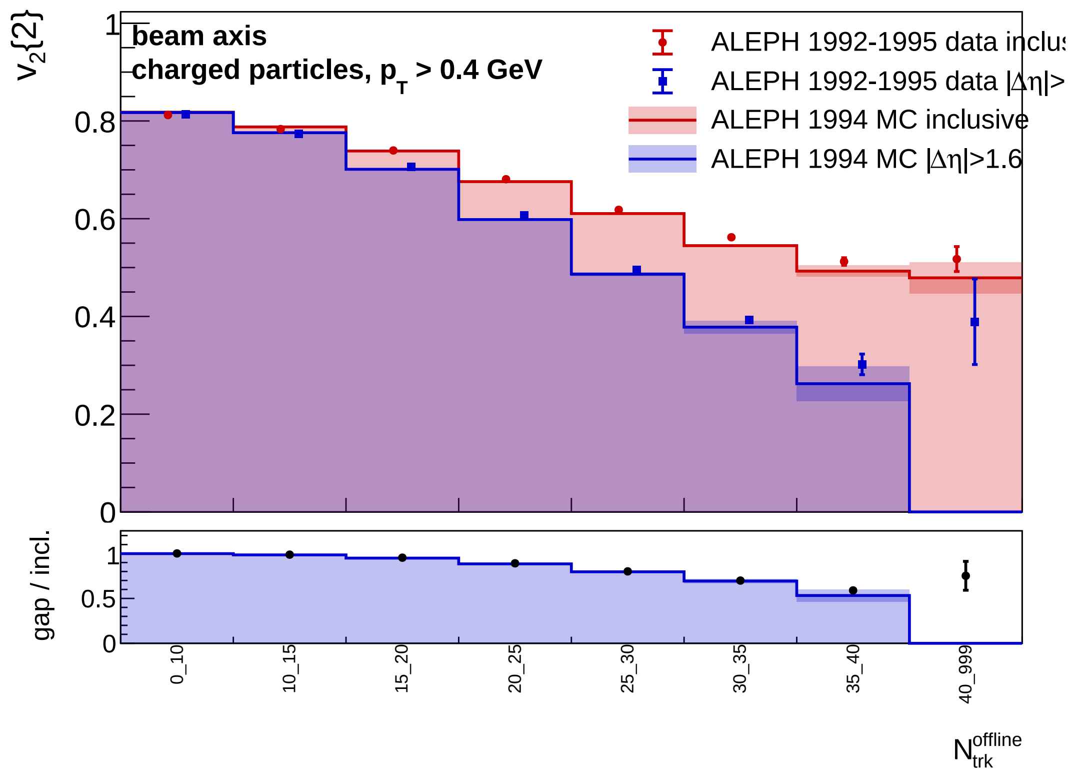



Figure 34: Thrust-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 2\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. The rapidity gap is evaluated using pseudorapidity around the thrust axis; the multiplicity bin remains the selected laboratory charged-particle multiplicity.

(a) Inclusive and rapidity-gap values




(b) Data/MC ratios

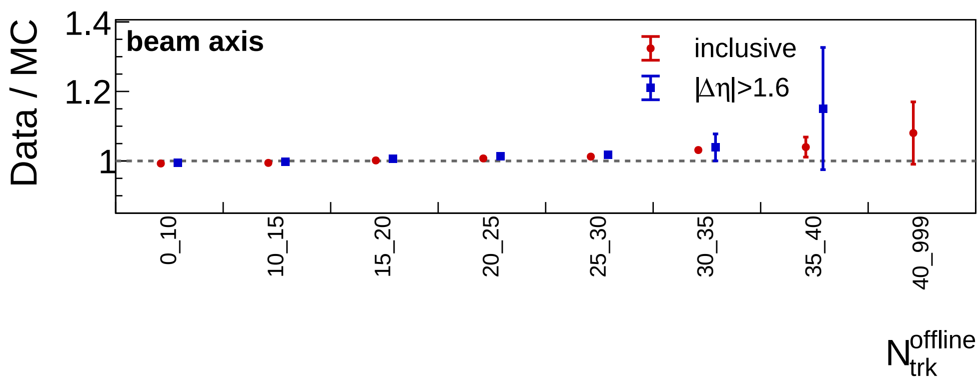
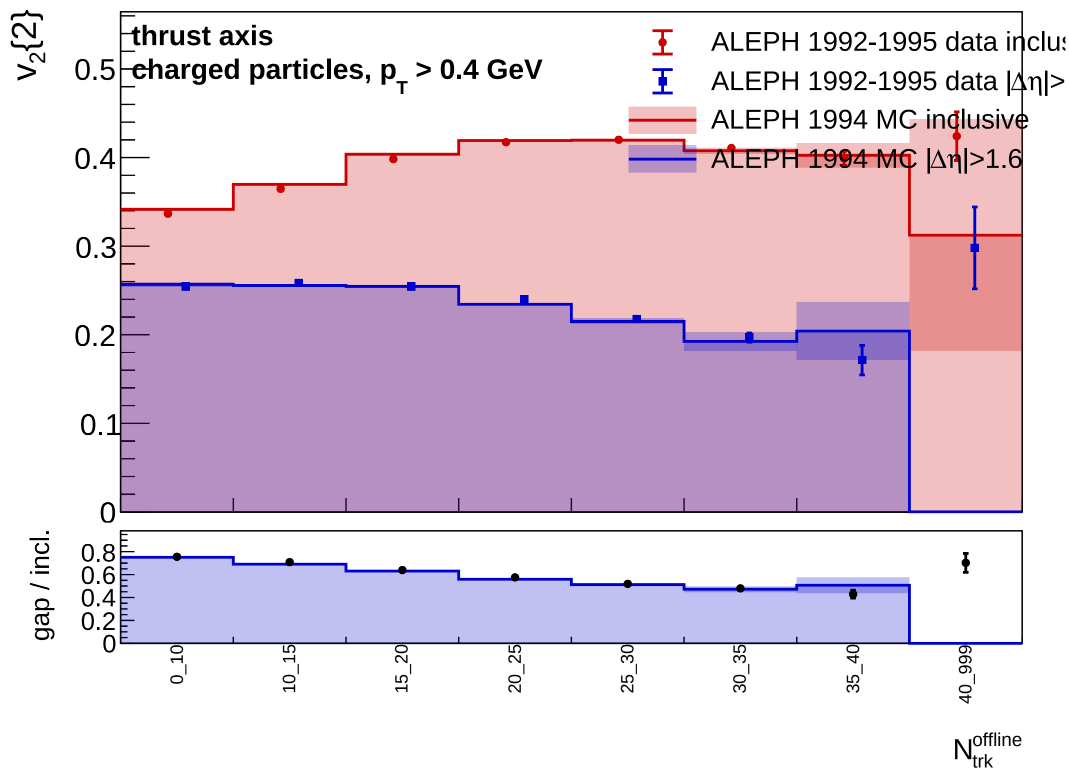



Figure 35: Beam-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 1.6\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. The lower subfigure shows the data/MC ratio for the inclusive and rapidity-gap observables with propagated statistical uncertainties.

(a) Inclusive and rapidity-gap values




(b) Data/MC ratios

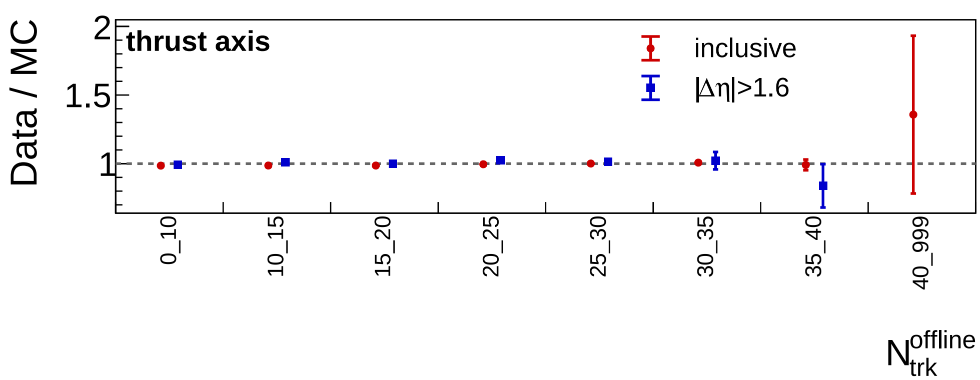



Figure 36: Thrust-axis comparison of inclusive $v_2\{2\}$ and $v_2\{2, |\Delta\eta| > 1.6\}$ for LEP1 1992–1995 data and 1994 reconstructed MC. The lower subfigure shows the data/MC ratio for the inclusive and rapidity-gap observables with propagated statistical uncertainties.

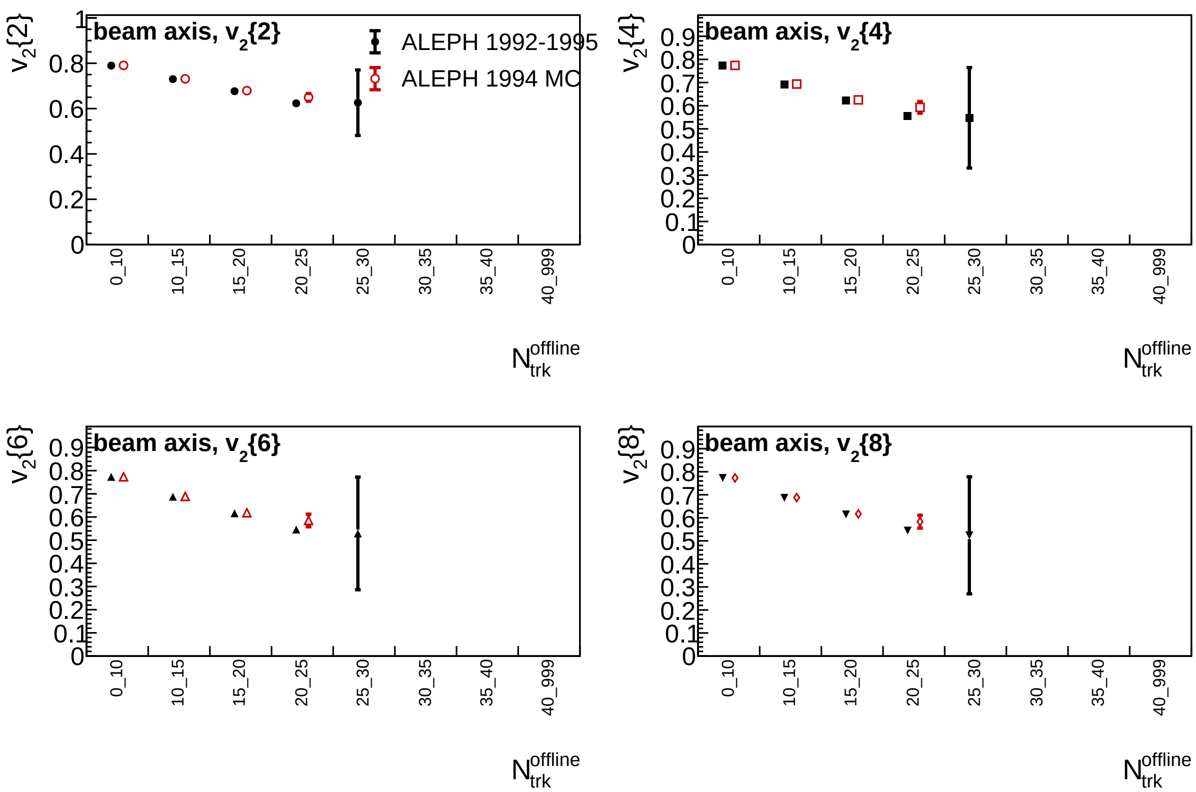



Figure 37: Beam-axis comparison of $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ for LEP1 1992–1995 data and 1994 reconstructed MC using selected charged particles with $1 < p_{\mathrm{T}} < 3$ GeV.

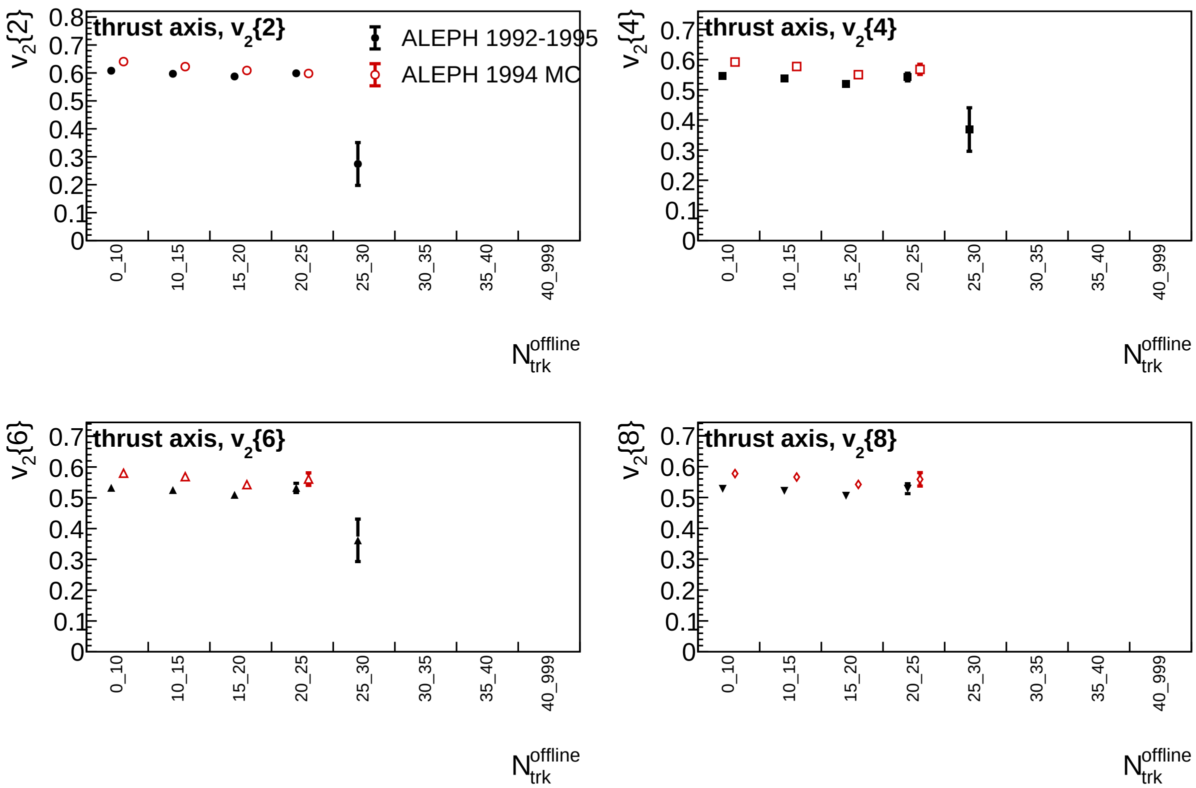
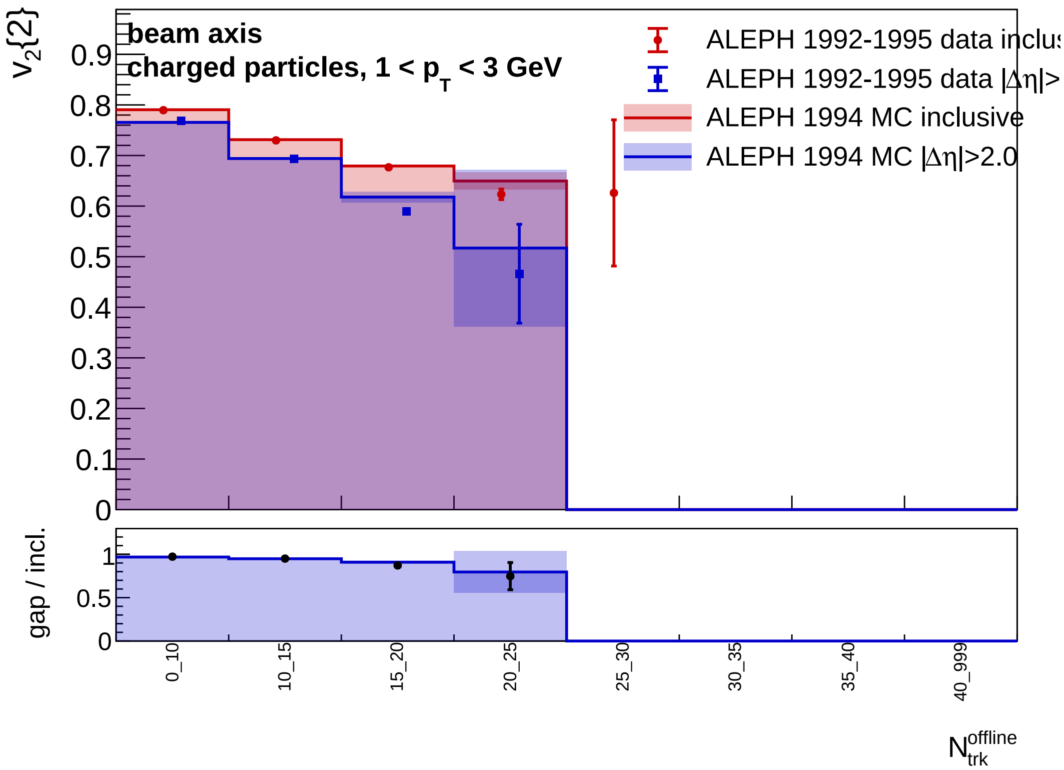



Figure 38: Thrust-axis comparison of $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ for LEP1 1992–1995 data and 1994 reconstructed MC using selected charged particles with $1 < p_{\mathrm{T}} < 3$ GeV, where $p_{\mathrm{T}}$ is evaluated relative to the thrust axis.

**(a)** Inclusive and rapidity-gap values




**(b)** Data/MC ratios

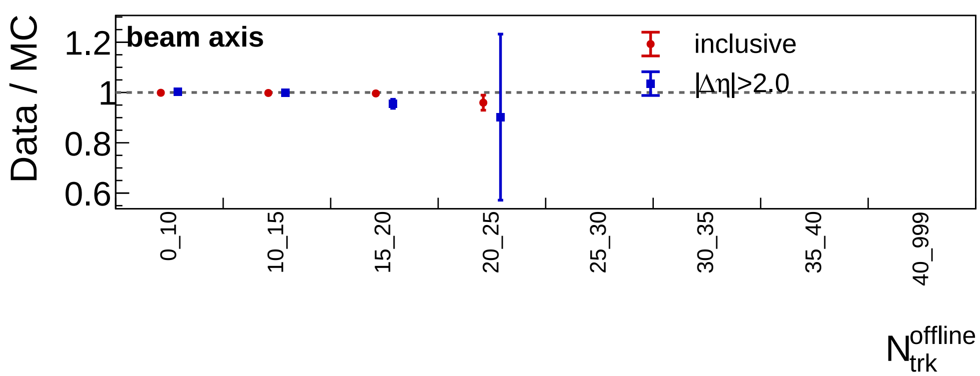
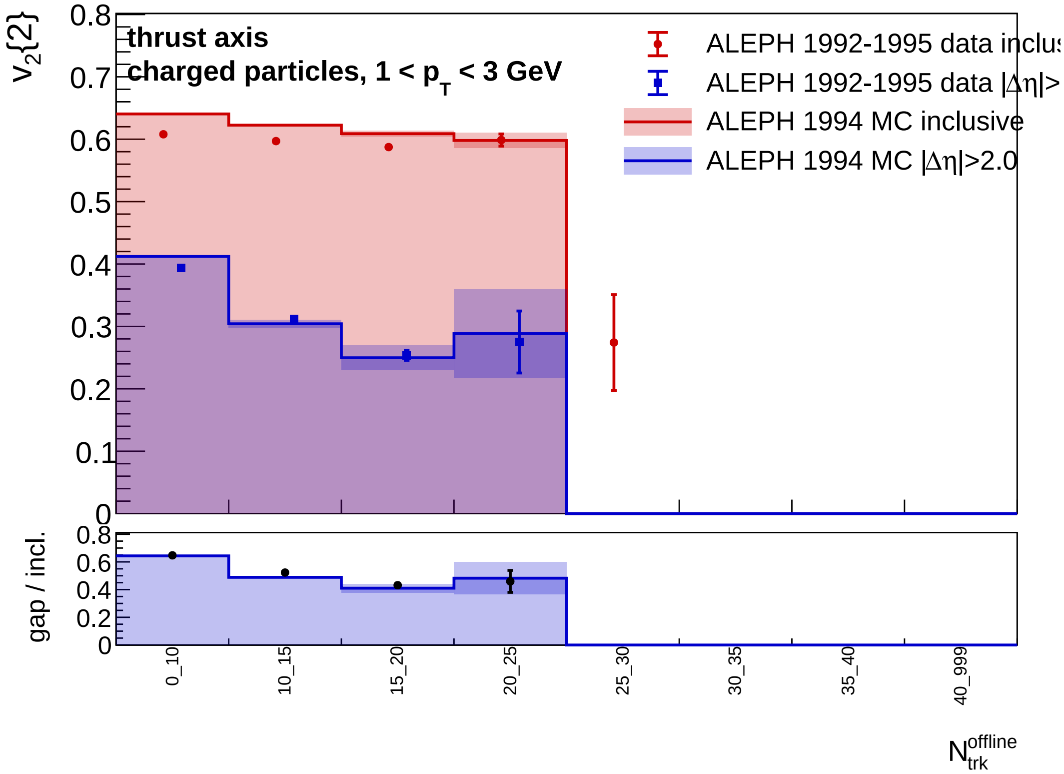



Figure 39: Beam-axis $v_2\{2\}$ comparison with $1 < p_\mathrm{T} < 3$ GeV selected charged particles. The rapidity-gap observable uses $|\Delta\eta| > 2.0$.

(a) Inclusive and rapidity-gap values




(b) Data/MC ratios

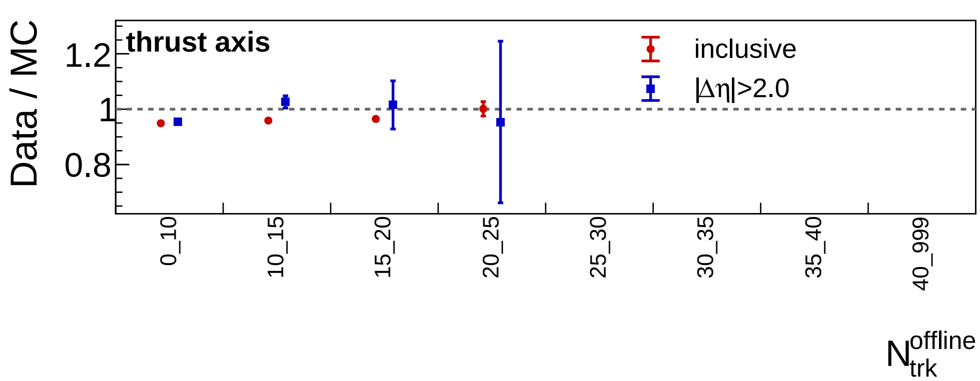
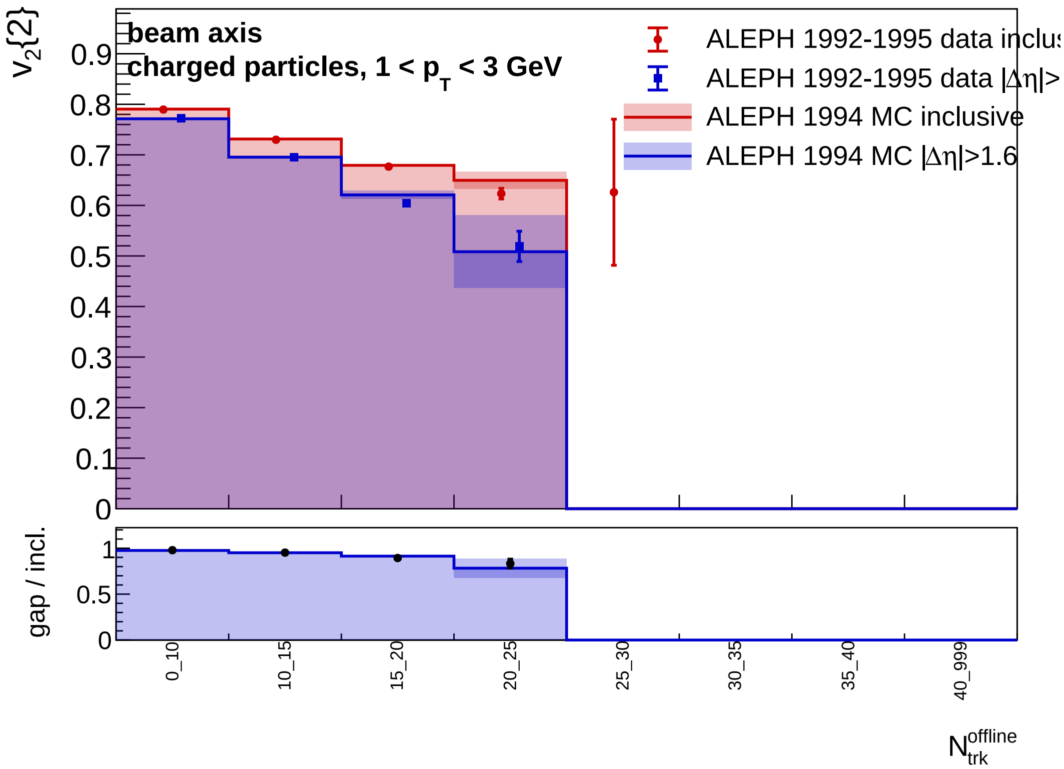



Figure 40: Thrust-axis $v_2\{2\}$ comparison with $1 < p_{\mathrm{T}} < 3$ GeV selected charged particles. The rapidity-gap observable uses $|\Delta\eta| > 2.0$ in the thrust-axis coordinate frame.

**(a)** Inclusive and rapidity-gap values




**(b)** Data/MC ratios

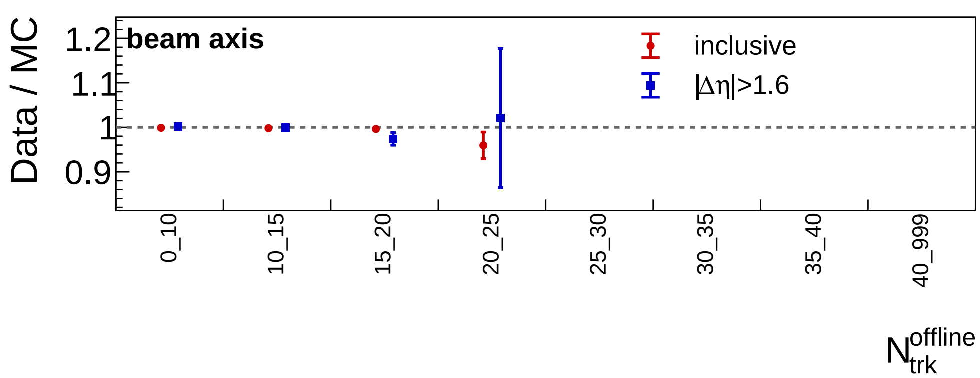
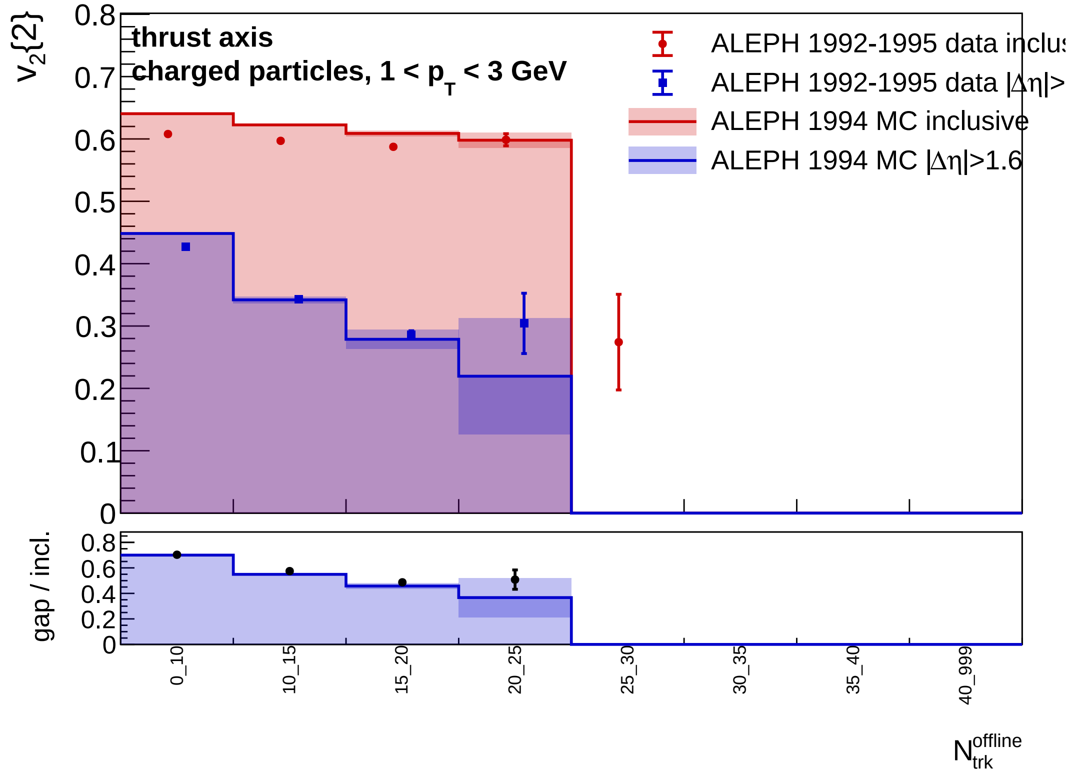



Figure 41: Beam-axis $v_2\{2\}$ comparison with $1 < p_{\mathrm{T}} < 3$ GeV selected charged particles and $|\Delta\eta| > 1.6$.

(a) Inclusive and rapidity-gap values




(b) Data/MC ratios

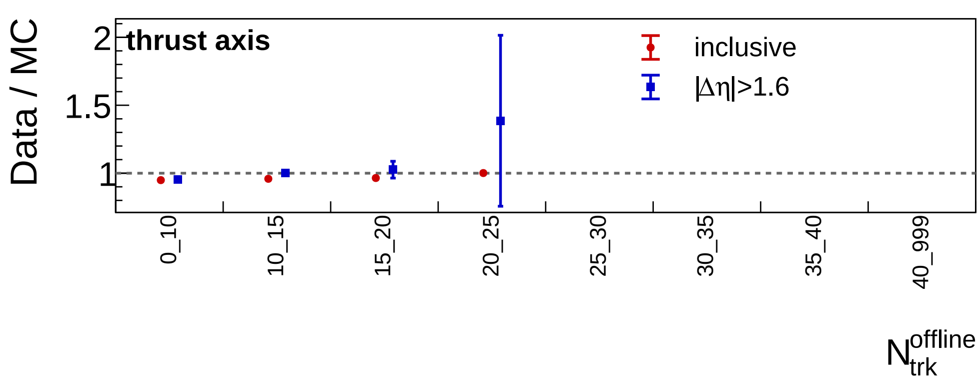



Figure 42: Thrust-axis $v_2\{2\}$ comparison with $1 < p_{\mathrm{T}} < 3$ GeV selected charged particles and $|\Delta\eta| > 1.6$ in the thrust-axis coordinate frame.

**Two-subevent cumulant definition.** For each axis frame, charged particles are split into subevent A with $\eta < -\eta_{\rm gap}$ and subevent B with $\eta > \eta_{\rm gap}$. The result shown here uses $\eta_{\rm gap} = 0$, i.e. a two-hemisphere split rather than a finite exclusion gap. The $2k$-particle correlation samples $k$ distinct particles from A and $k$ distinct particles from B,

$$\langle 2k \rangle_{\rm 2sub} = \left\langle e^{in(\phi_1^A + \cdots + \phi_k^A - \phi_1^B - \cdots - \phi_k^B)} \right\rangle,$$

with ordered tuple weights $M_A(M_A-1)\cdots(M_A-k+1)\; M_B(M_B-1)\cdots(M_B-k+1)$. The same cumulant combinations used for the all-particle result are then applied to $\langle 2\rangle_{\rm 2sub}$, $\langle 4\rangle_{\rm 2sub}$, $\langle 6\rangle_{\rm 2sub}$, and $\langle 8\rangle_{\rm 2sub}$. $v_2\{2k\}_{\rm 2sub}$ is plotted only when the corresponding cumulant has the real-valued sign convention.

### 6.8 Two-subevent detector-level data/MC comparison

The LEP1 1992–1995 data and 1994 reconstructed-MC samples were reprocessed with the same track selection, multiplicity bins, and sixteen-chunk jackknife procedure, adding the two-subevent cumulant sums described above. Figures 43 and 44 show MC as histograms with statistical bands and data as points with jackknife uncertainties.

For the beam axis, the two-subevent result is stable and the data/MC ratio is near unity over the populated range. The ratio is about 0.988–1.016 for $v_2\{2\}_{\rm 2sub}$ through 20–25 charged tracks and remains within the larger statistical uncertainties in the high-multiplicity tail. The higher-order beam-axis cumulants show the same qualitative behavior, with increasing statistical spread above 30 charged tracks.

For the thrust axis, $v_2\{2\}_{\rm 2sub}$ is broadly reproduced by the MC in intermediate multiplicity bins, but the higher-order two-subevent cumulants are statistically fragile. Several bins are sign-invalid in either data or MC, and valid high-order points have large jackknife uncertainties. These thrust-axis higher-order points should therefore be treated as a diagnostic of the current sample size and subevent definition, not as a corrected physics comparison.

## 7 Current Limitations

The present note should be read as a detector-level analysis milestone, not as a final corrected physics result. The main limitations are:

- no detector-response or acceptance correction is applied to the cumulants;
- the current reconstructed-MC comparison is a baseline check, not a detector-response correction or generator systematic uncertainty;
- only one track-$p_{\rm T}$ window variation, $1 < p_{\rm T} < 3$ GeV, is included so far; broader charged-particle definition, thrust-axis construction, and event-selection variations are still needed;
- the LEP2 merged-sample cross-check currently uses the requested full `LEP2_merged.root` file without an additional energy-window selection;
- high-order cumulants in the highest-multiplicity bins are statistically fragile and can become sign-invalid.

These items define the next analysis steps before the result can be interpreted as a corrected ALEPH measurement.

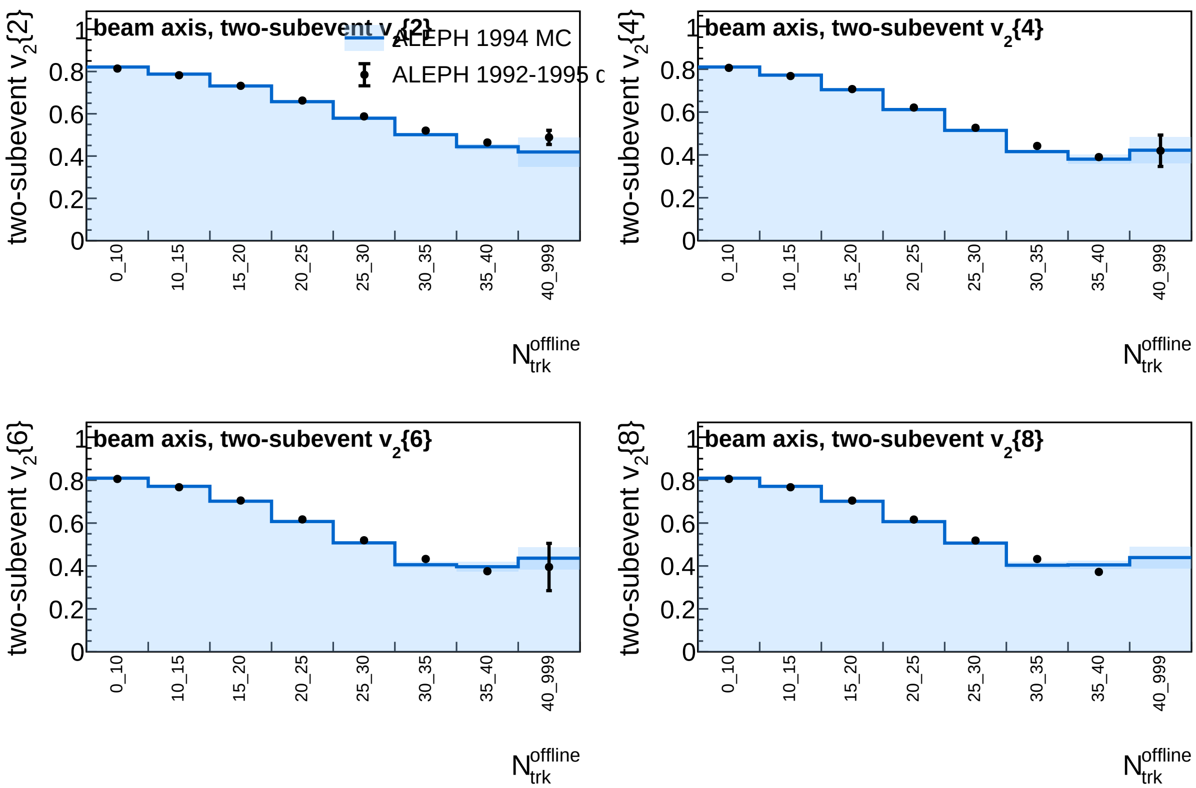



Figure 43: Beam-axis two-subevent comparison of LEP1 1992–1995 data and 1994 reconstructed MC. MC is drawn as a histogram with statistical band, while data are shown as points with delete-one-chunk jackknife uncertainties.

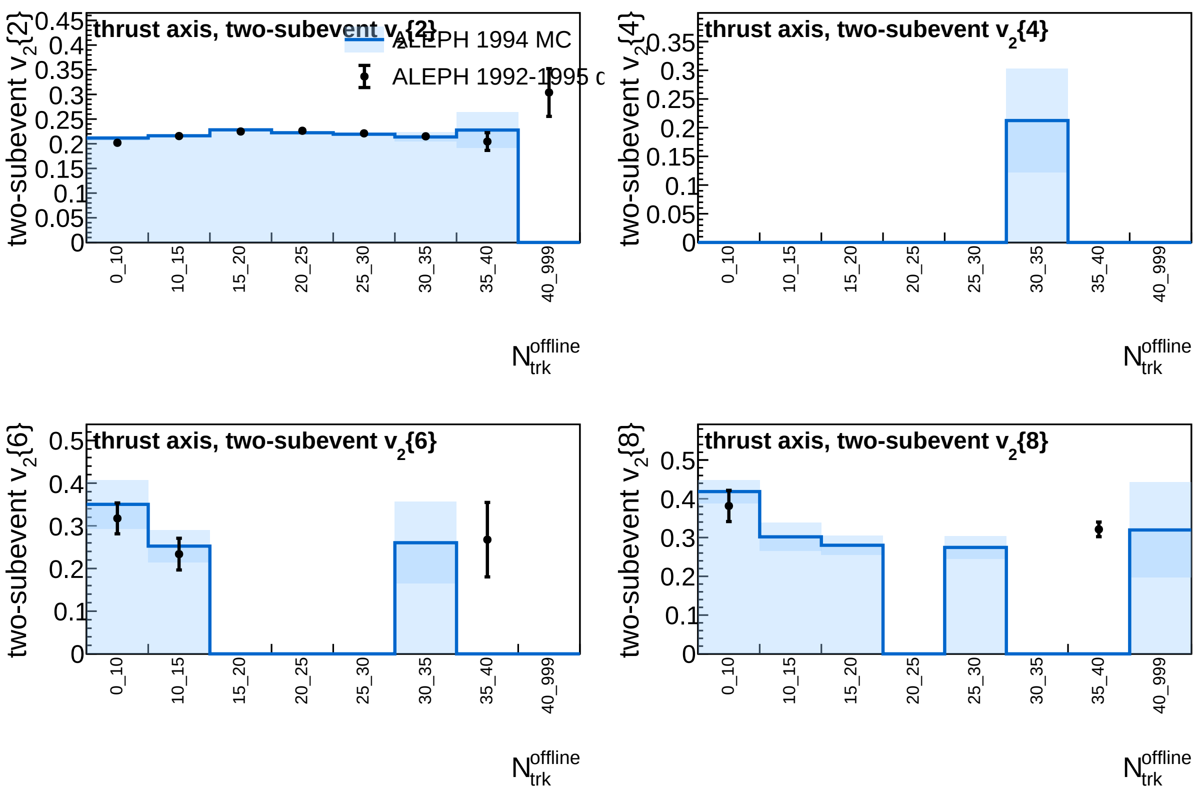



Figure 44: Thrust-axis two-subevent comparison of LEP1 1992–1995 data and 1994 reconstructed MC. Missing or zero high-order entries correspond to sign-invalid cumulants or bins without a valid real-valued $v_2\{2k\}_{2\mathrm{sub}}$.

# 8 Summary

A first ALEPH charged-particle cumulant workflow has been built from the local StudyMult code base and validated against exact brute-force ordered-particle sums. The current figures use the merged LEP1 1992–1995 reconstructed data sample, with 3,050,610 selected events in the analysis multiplicity bins. The analysis extracts $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus charged-particle multiplicity for both the beam-axis and thrust-axis azimuthal definitions, and compares inclusive $v_2\{2\}$ with rapidity-gap selections $|\Delta\eta| > 2.0$ and $|\Delta\eta| > 1.6$. An additional detector-level cross-check repeats the inclusive and rapidity-gap measurements for charged particles with $1 < p_{\mathrm{T}} < 3$ GeV. A first LEP2 merged-sample data cross-check has also been produced from the requested `LEP2_merged.root` input file. Statistical uncertainties are propagated through the nonlinear cumulant definitions with a sixteen-chunk jackknife. The beam-axis and thrust-axis figures included in this note provide the current detector-level baseline for subsequent correction, 1991-data recovery, and systematic-uncertainty studies.

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